

Rulemakers and Ruletakers in Digital Trade Governance: Network Structure and the Diffusion of Digital Provisions in Preferential Trade Agreements

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Abstract

Who makes the rules of digital trade governance, and how do those rules spread? As multilateral negotiations stall, countries increasingly embed digital governance rules in preferential trade agreements (PTAs), creating a fragmented regime complex of overlapping and non-hierarchical commitments. Leveraging the TAPED dataset, I build three provision-type networks (e-commerce, data flows, and intellectual property) and analyze 382 PTAs encompassing 173 countries (2000-2022) through descriptive network analysis and Stochastic Actor-Oriented Models (SAOMs). I propose a novel five-category country typology, showing that structural centrality alone is a poor proxy for rulemaking agency: Peru and Vietnam rank higher than the United States in terms of centrality, yet neither designs the provisions they adopt. The SAOM results showcase that e-commerce and data governance networks grow through a structural arbitrage mechanism rather than generic clustering, while the intellectual property network follows a qualitatively different logic consistent with the influence of TRIPS-Plus. Formation dynamics vary across provision types: data governance is the most hierarchy-driven domain, while IP provisions diffuse largely independently of economic capacity. The EU's collective negotiation structure produces a distinctive pattern of external regulatory export while imposing regulatory constraints without providing corresponding market-access provisions. These findings suggest that the distinction between those who set the rules of digital governance and those who adopt them is shaped by the interplay of entry timing, structural constraints, and the specific provisions at stake.

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1 Introduction

When Peru, Vietnam, and Mexico joined the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP) in 2018, they adopted a digital trade chapter prohibiting data localization requirements, protecting source code from forced disclosure, and mandating free cross-border data flows. These provisions built on templates that first appeared in the US-Singapore FTA of 2004, among the earliest PTAs with a comprehensive digital trade chapter (Burri and Polanco 2020; Elsig and Klotz 2022), and pushed for the most part by both these countries (Jones et al. 2024). Meanwhile, the EU-Vietnam Free Trade Agreement (2020) embedded an entirely different regulatory model: data protection standards, consumer privacy requirements, and regulatory cooperation obligations reflecting the European approach to digital governance (Elsig and Klotz 2021a; Bradford 2023). Vietnam itself now participates in both regulatory models through different agreements, having effectively implemented different sets of digital governance standards through distinct trade policy channels.

These events showcase two different aspects related to digital governance: powerful, interested countries have the tools to shape the digital trade rules that other countries adopt, and trade policy is undoubtedly one of the most prominent channels through which these rules can be standardized at the international level. Indeed, while digital governance is increasingly becoming one of the most important regulatory domains of the twenty first century, sustaining digital integration requires coordinated regulation that no current multilateral framework provides (Weymouth 2023). The WTO Work Programme on Electronic Commerce, despite updates and renewed discussions, has still relatively loose rules and a limited scope: the Joint Statement Initiative of 2019, for example, yielded a thin set of provisions that focused on trade facilitation rather than expanding significantly the regulatory scope of the WTO (Burri 2022). Preferential Trade Agreements (PTAs) have thus become the policy tools of choice for countries to coordinate their digital governance frameworks, creating an institutional *regime complex* of overlapping and non-hierarchical agreements (Raustiala and Victor 2004; Alter and Meunier 2009) governing matters related to digital trade, privacy, and intellectual property rights (IPR). This paper aims to focus on this regime complex that en-

compasses over four hundred PTAs containing digital provisions of varying scope and depth, from customs duty prohibitions to comprehensive data protection obligations and source code protections (Burri and Polanco 2020). Focusing on this specific regulatory domain rather than multilateral bodies (WTO, ITU, ICANN)¹ and unilateral regulation (GDPR, China’s PIPL)², I aim to answer a broader question: *Through what structural mechanisms do digital governance rules spread across the PTA network, and what distinguishes countries that design these rules from those that merely adopt them?*

Scholarship in international political economy and international law has made substantial progress on both fronts, suggesting that competing regulatory frameworks appear to be backed by the most influential states (Aaronson and Leblond 2018; Bradford 2023), and that, while digital integration demands coordinated regulation, the multilateral system has failed to provide it (Weymouth 2023). Provision-level mappings have traced how these models travel across agreements (Elsig and Klotz 2021a, 2022), often through verbatim template copying (Allee and Lugg 2016; Allee et al. 2017a; Peacock et al. 2019; Allee and Elsig 2019) and sometimes through deliberate norm entrepreneurship by countries that design agreements as vehicles for broader adoption (Jones et al. 2024). A related line of studies has shown that asymmetric positions in global networks provide coercive leverage on hub states (Farrell and Newman 2019), and that network-based mechanisms appear to apply both to PTAs in general and to specific Non-Trade provisions (Milewicz et al. 2018), raising the question of whether similar dynamics might shape who sets rules in digital governance. Yet, these two bodies of work, one mapping content and diffusion, the other theorizing network power, have not been systematically connected. We know how digital provisions spread across agreements and we understand the importance of network structure, but we know far less about how network position shapes the diffusion of digital governance rules and distinguishes countries that design those rules from those that end up just adopting them.

¹In order, the World Trade Organization; The International Telecommunication Union, the United Nations agency for information and communication technologies; and the Internet Corporation for Assigned Names and Numbers, which coordinates the internet’s domain-name system.

²In order, the European Union’s General Data Protection Regulation (2016), referred to as the benchmark, most-comprehensive cross-border data-protection regime; and China’s Personal Information Protection Law (2021), its principal personal-data-protection statute.

I argue that the distinction between rulemakers and ruletakers in digital governance is produced by the interaction of three forces. The first is entry timing: countries that signed the earliest digital trade agreements (the US-Jordan FTA of 2000, the US-Singapore FTA of 2004, the Trans-Pacific Strategic Economic Partnership of 2005) were writing provisions that did not yet exist elsewhere, while latecomers adopted templates already in circulation. The second is structural arbitrage: when a hub country signs separate agreements with two partners, those partners face pressure to form a direct tie that cuts the hub’s intermediary position, carrying the hub’s provision templates outward in the process (Manger et al. 2012). The third is provision-type heterogeneity: because e-commerce, data flow, and IP provisions are substantively different in terms of regulatory effects and institutional origins, these dynamics do not operate uniformly across the three networks. Taken together, these forces produce distinct governance environments in which structural centrality is a poor proxy for rulemaking agency; countries like Peru and Vietnam can gain central network positions through mega-regional agreements without ever having designed the provisions they adopt.

To evaluate these claims, I construct three separate PTA networks covering e-commerce, data-flow, and intellectual-property provisions using the TAPED dataset (Burri and Polanco 2020). I characterize the structural properties of each network, propose a country typology that distinguishes rulemakers from adopters, and estimate Stochastic Actor-Oriented Models (SAOMs) to identify the mechanisms driving tie formation across provision types. This approach extends existing network analyses of PTA formation (Manger et al. 2012) and non-trade agenda diffusion (Milewicz et al. 2018) to the domain of digital governance.

The analysis yields three main findings. First, using a provision-level novelty score, I propose a typology of four measured roles (Rulemakers, Adopters, Brokers, and Peripheral) plus the EU as a structurally distinct bloc, identifying countries whose network position is consistent with template design versus those whose centrality is borrowed from agreements designed by others. Second, e-commerce and data governance networks appear to grow through the closure of indirect connections rather than generic clustering, while the IP network follows a qualitatively different logic. Third, provision types appear to diffuse through different means: while data governance is the most hierarchy-driven domain, with economic factors

strongly predicting tie formation, IP provisions diffuse regardless of economic capacity. These findings are also helpful in differentiating actors that shape digital governance from those that primarily participate in it. The European Union emerges as a distinctive regulatory exporter, while the United States retains the rulemaking role long emphasized in the literature. China, by contrast, occupies a central bridging position yet authors relatively few of the provisions that circulate through the network, a finding that challenges accounts portraying China as a co-equal leader of global digital governance alongside Washington and Brussels (Bradford 2023).

The paper is organized as follows. Section 2 develops the theoretical framework, Section 3 describes the data, network construction, and the analytical methodology, Section 4 presents results relating to country roles and the SAOM analysis, Section 5 provides a discussion of the findings, and Section 6 concludes.

2 Digital Governance Through Trade Agreements

2.1 Digital Regionalism: The Drivers of Digital Trade Globalization

The shift from multilateral to preferential governance of digital trade can be seen both as a failure of the broader multilateral consensus, and as a strategic opportunity for many actors willing to create new sets of rules. Indeed, even the WTO’s more recent take on digital governance (the Joint Statement Initiative on E-commerce) has been criticized for its limited scope and lack of ambition, while comprehensive digital rules have been proliferating through PTAs since the early 2000s (Elsig and Klotz 2021b; Burri 2022; Weymouth 2023). This potential cross-institutional shift leads to what Alter and Meunier (2009) describe as “chessboard politics”, a regime complex of parallel and overlapping institutions in which actors select the environments that best allow them to pursue their policy preferences. Across the international system, complexity and fragmentation have been increasing in different areas of global governance (Alter and Raustiala 2018), and both trade policy and digital governance are no exception: overlapping agreements with different standards

and different sets of actors have been proliferating, creating a complex web of rules and institutions that govern digital trade. New kinds of provisions such as data localization, algorithmic transparency, and source code protection emerge faster than multilateral treaties can absorb them, forcing states to seek bilateral and plurilateral channels to advance their policy objectives and rulemaking ambitions. PTAs offer a platform for experimentation and evidence-gathering that the multilateral system cannot replicate (Burri 2022), fostering a fragmented governance order that ends up being deeper, more specific and more varied than anything laid out in the WTO.

Among these fragmented arrangements, three main regulatory models have been identified as the most influential: a market-driven, a rights-driven, and a state-driven system, all in competition with one another as the US, China and the EU seek to export their target models (Elsig and Klotz 2021a; Bradford 2023). Even without entering the digital policy sphere, Horn et al. (2010) documented the main differences between European and American PTA models, with EU agreements containing nearly four times as many provisions going beyond existing WTO commitments, though much of this coverage takes the form of “legal inflation” where provisions are broad in scope but limited in enforceability. The US, by contrast, includes fewer provisions but makes them legally binding, particularly in labor and environmental standards. These differences stem from domestic policy choices, and both actors have shown clear ambitions to export their regulatory models abroad. Dür et al. (2014) underscore some of these broader distinctions as well as the fact that aggregate depth might hide some of the content-level variation across PTA chapters, warranting more provision-level disaggregation.

Given how the existence of distinct regulatory models is established, the question becomes how do these systems diffuse. Simmons et al. (2006) identify four mechanisms of international policy diffusion: coercion, competition, learning, and emulation. In the PTA context, competitive dynamics are the most direct channel, but all of these mechanisms appear to be relevant. Baldwin and Jaimovich (2012) show that trade diversion pressure generates contagion effects, as countries excluded from a new agreement rush to form their own PTAs to avoid being left behind; Elkins et al. (2006) document a parallel process in investment

treaties, where competition for capital drove a series of bilateral agreements that accelerated through the 1990s. Beyond competitive pressure, a large share of provisions gets simply copied verbatim: over half of PTA content is reproduced from prior agreements through some of the same templates (Allee and Elsig 2019), using “boilerplate” language transplanted across agreements with minimal adaptation (Peacock et al. 2019). This phenomenon also applies to more institutional settings, where the same mechanism operates when countries adopt WTO language into bilateral PTAs (Allee et al. 2017b). Preferred rules are thus often pushed by powerful economic actors (Drezner 2008; Allee and Lugg 2016), but once set these rules will also influence other states who will inherit the design choices embedded in previous templates. Jones (2024) takes the diffusion argument a step further, showing that Singapore designed the Digital Economy Partnership Agreement (DEPA), a modular agreement intended to be adopted and replicated by other countries, to set itself as “norm entrepreneur” in the digital governance arena. This phenomenon is not rare, and related to the fact that “legal language has a way of sticking around” (Castle 2023). Taken together, these diffusion processes can imply that a country can become structurally central in a governance network without ever having designed the provisions it abides by. Template adoption, competitive emulation, and boilerplate copying all allow states to accumulate ties through rules written by other actors, which means that structural centrality in a provision network is not, by itself, always evidence of rulemaking agency.

Agency is not the only relevant component, though: not all digital governance rules face the same regulatory demands, institutional constraints, or political pressures, and these differences have consequences for how provisions spread across agreements. The provision *type* thus matters.

E-commerce provisions, which relate to digital products, customs duty prohibitions on electronic transmissions, and electronic transaction frameworks, represent the most mature domain. Built on the WTO’s 1998 moratorium on customs duties for electronic transmissions and reinforced through two decades of PTA practice, these provisions impose relatively low regulatory demands: they codify principles that are already widely accepted, and the templates for doing so are readily available (Burri and Polanco 2020; Elsig and Klotz 2021b).

Data flow provisions position themselves somewhat differently: rules governing cross-border data transfers, data localization prohibitions, and personal data protection lack a true multilateral baseline, suggesting that PTA-based templates have had more room for policy flexibility and design choices linked to data flows (Aaronson and Leblond 2018; Elsig and Klotz 2021a).

IP provisions, dealing with source code protection, technology transfer requirements, and IP enforcement mechanisms, sit at the other end of the institutional spectrum. Shaped by the TRIPS Agreement and its successive extensions, IP rules in PTAs build on and extend an existing multilateral framework (known as TRIPS-Plus) rather than creating one from scratch. The sharpest contrast stands between IP and data governance: while IP provisions can draw on decades of codified multilateral language, data flow rules are being written in real time through bilateral and plurilateral negotiation, with competing models backed by different actors and coalitions of states.

Prior research has identified bilateral and country-level factors (GDP, geographic distance, domestic regulatory frameworks...) that predict whether a given pair of countries will include digital provisions in their trade agreements (Peressotti 2025). What remains unaddressed are the network-level dynamics that shape who ends up at the center of this governance architecture and who remains at its periphery. The question shifts from a dyadic one (what makes two countries include digital provisions in a shared PTA?) to a more structural one: through what network-level mechanisms do some countries become *rulemakers* while others become *ruletakers*?

If provision types differ in their regulatory demands and institutional origins, the network dynamics governing their diffusion should differ as well. E-commerce provisions build on two decades of WTO practice (the 1998 moratorium on customs duties for electronic transmissions and the Trade Facilitation Agreement), and the templates for codifying them are widely available (Burri and Polanco 2020; Elsig and Klotz 2021b). In this instance, since a multilateral set of rules already exists, the asymmetry between template designers and template adopters should be weakest. Data flow would land at the opposite side of the spectrum: rules governing cross-border data transfers and data localization prohibitions lack

a multilateral starting point, which means that a template design has more room for the introduction of distinct provisions. Structural arbitrage should be strongest in this domain, because the absence of ready-made language raises the value of a hub’s existing templates. IP provisions sit in an intermediate institutional position: shaped by the TRIPS Agreement and its successive extensions, they build on a multilateral framework rather than creating one from scratch. Provision-type heterogeneity thus affects digital governance, suggesting that e-commerce, data flows, and IP provisions will exhibit distinct network formation dynamics.

Based on the arguments I just laid out, if three competing regulatory models (Bradford 2023) and distinct “data realms” (Aaronson and Leblond 2018) structure digital governance, specific network communities should cluster by regulatory ideology rather than geography: the US-led market model, the EU’s rights-based framework, and China’s state-driven approach should each generate identifiable community structure. The first-mover advantage documented by Elsig and Klotz (2022) should translate into durable structural positions: the earliest entrants into each provision network should remain the most central actors in 2022. If network centrality captures rulemaking agency as Farrell and Newman (2019) suggest, the most central countries should also be those whose regulatory templates other actors end up adopting. Yet mega-regional agreements like the CPTPP and RCEP may inflate the centrality of countries that joined large coalitions without shaping their content, which means centrality alone would not reliably distinguish rulemakers from rule takers. The country typology I propose in Section 3 addresses this by combining centrality measures with entry timing and structural position to classify countries into distinct governance roles.

2.2 Network Structure and Digital Rulemaking

Network analysis can help to tie together the role that different actors (states) have on defining digital governance through domestic, bilateral, or multilateral action. And indeed, network research has already established that bilateral relationships are not independent: existing connections shape new tie formation through transitivity, preferential attachment, and homophily (Hafner-Burton et al. 2009). Farrell and Newman (2019) push this insight further, showing that asymmetric network structures can generate power imbalances; states

with jurisdiction over network hubs can exploit those positions for surveillance and coercion. Castle (2023) makes a related argument through strategic sequencing: states use early, low stakes agreements to set precedent, and the consequent advantages are reaped from those with the legal capacity to shape the templates others later adopt. If hub states can weaponize network position in financial and information infrastructure, similar dynamics may also shape who sets rules in trade agreement networks. Kim and Manger (2017) further advance this argument for PTAs, showing that hub countries create governance templates that have the potential to lock in subsequent partners through path dependence. Once a hub selects a particular regulatory design (such as a positive-list or negative-list approach to services, say, or a specific data localization framework) that choice propagates through the network as new partners adopt the hub’s template rather than negotiating from scratch. In digital governance, such path dependence is visible in Singapore’s and the United States’ early template choices, which have cascaded through subsequent agreements as if constraining the governance options available to latecomers. For PTA networks more broadly, Manger et al. (2012) also show, using SAOMs, that spoke countries in hub-and-spoke configurations form  direct ties to eliminate the hub’s structural advantage, a dynamic they term *structural arbitrage*: their finding that richer countries (in a form of income hierarchy) shape PTA formation provides one of the conceptual foundations for this analysis. Milewicz et al. (2018) extend the framework to non-trade issues, showing that nontrade issue adoption follows a similar network diffusion logic through mechanisms of network-mediated learning and competitive pressure (see also Kinne 2013).

I argue that structural arbitrage extends to provision-specific digital provisions, and that it has concrete implications for digital governance networks as well. Consider a specific network process: when a hub country (say, Singapore) signs separate PTAs with two spoke countries (say, Australia and Chile), Singapore becomes the conduit through which Australian and Chilean firms access each other’s digital markets. Australia and Chile would then face a choice: remain dependent on Singapore as intermediary, or form a direct tie that eliminates this structural dependence. The structural arbitrage hypothesis predicts they will choose the latter, especially when digital provision templates are already available from prior agreements and negotiation costs are correspondingly low (Allee and Elsig 2019; Jones et al. 2024). In

the context of SAOM modeling, this process is captured by the “number of pairs at distance 2” effect: countries with many indirect connections (partners of partners with whom they lack a direct tie) should experience structural pressure to close those gaps³. If this mechanism is the primary driver of triangle formation (three actors each having a connection with one another), then controlling for it should make generic triadic closure non-significant: countries form triangles not because they prefer clustering per se, but because they are eliminating indirect structural dependence. To clarify: under generic closure, any two countries with a shared partner face pressure to connect; under structural arbitrage, only the two spokes have an incentive to form a direct tie, because the hub actually prefers the triangle to stay open. If spoke-specific incentives do not govern tie formation, triangle formation should look no different from ordinary clustering. This leads to my first hypothesis:

Hypothesis 1 (*Structural arbitrage*): Countries connected indirectly through a common hub should form direct ties to eliminate the hub’s intermediary advantage.

The position states hold in a network is not only a function of structural dynamics (such as centrality); it also accounts for other relational and actor-level characteristics like different economic attributes. Wealthier countries possess both the demand for governance provisions, given the size of their digital economies, and the negotiating resources to draft complex chapters. In network terms, economic capacity should generate degree asymmetry: high-GDP countries accumulate more ties, producing the hub-and-spoke configurations that then give their partners something to arbitrage around. The same institutional ordering that differentiates the three networks sets how steep that hierarchy runs. It should be sharpest in data flow, where no multilateral floor exists and the wealthy countries with the capacity to draft complex chapters dominate template design (Aaronson and Leblond 2018); weakest in IP, where ready-made TRIPS-Plus language lets even lower-capacity states take on commitments without leaning on a wealthy partner’s template; and intermediate in e-commerce, whose provisions are widely standardized yet still demanding in their most ambitious chap-

³A pair at distance 2 refers to two countries connected through a single intermediary but not directly linked to one another. Unlike triangles, which capture closed triads, this effect captures the presence of indirect relationships that can potentially be closed through future tie formation.

ters on electronic authentication, online consumer protection, and paperless trading. As Elkins et al. (2006) show in the investment-treaty context, wealthier countries set the competitive benchmark others must match; in digital governance that benchmark is the design of the provisions themselves, which is precisely why the asymmetry should be more obvious where the provisions are newest.

Economic similarity should reinforce these dynamics. Countries at comparable income levels face similar data-flow volumes, comparable regulatory infrastructure, and analogous positions vis-a-vis the US, EU, and Chinese models, which makes peer-to-peer coordination more likely precisely in the domains where rules are still being negotiated rather than inherited. Similarity should therefore promote ties in e-commerce and data governance, where peers converge on shared templates. In the IP network the logic would differ: asymmetric North-South TRIPS-Plus dynamics run between unlike partners, so income similarity should do little to explain who forms IP ties. I thus propose the following related hypotheses:

Hypothesis 2 (*Economic hierarchy*): Higher-GDP countries should be more active in forming digital governance ties, with hierarchy effects strongest in the data flow network and weakest in the IP network.

Hypothesis 3 (*Economic similarity*): Economic similarity should promote tie formation in e-commerce and data governance but not in the IP network.

Finally, the European Union represents a distinct case. EU member states negotiate trade agreements collectively through the European Commission and cannot form internal PTAs with one another, which channels all EU trade policy activity outward. The expectation of a distinctive EU network signature draws on three established findings. Bradford (2020) argues that the EU exports its regulatory standards, particularly data protection and consumer privacy, to trade partners that seek access to the European market, a dynamic she terms the Brussels Effect. Horn et al. (2010) documented this export pattern, showing that EU PTAs are far broader in regulatory scope than their US counterparts, even if much of this coverage is aspirational rather than legally enforceable: this would be a form of “declaratory diplomacy” that tracks the Brussels Effect logic. If this regulatory export operates through PTAs, the EU’s institutional constraint should produce a distinctive structure in digital

governance networks, with the strongest external orientation in areas where EU regulatory standards are most harmonized internally. The EU’s signature in these networks is therefore structural, not simply a matter of who exports the most: the United States also diffuses its templates, but through deep bilateral agreements that build little network breadth, so it does not produce the outward-only, regulation-heavy footprint that collective negotiation gives the EU. The SAOM models capture this through the “both EU members” homophily effect.

Hypothesis 4 (*EU structural constraint*): EU member states should form digital governance ties predominantly with non-EU partners, with external orientation intensifying from e-commerce to data to IP.

Taken together, these broad four hypotheses outline a framework in which the PTA-driven side of digital governance is not one phenomenon but a set of structurally distinct processes, shaped by economic hierarchy, mediated by institutional constraints, and embedded in network architectures that reward early movers and hub countries. In a way, the outline aims to test a broader, geopolitical theory connected to the claims that the US, the EU, and China are seeking to export competing regulatory visions (Aaronson and Leblond 2018; Bradford 2023) while expanding the known model of “international trade network” (Bhattacharya et al. 2007; Fagiolo 2010) beyond import-export ties. Aaronson and Leblond (2018)’s “data realms” and Bradford (2023)’s “digital empires” frameworks represent an ideological clash, leaving aside how template diffusion, structural arbitrage, and path dependence channel that competition into specific networks that advantage some actors over others. A purely geopolitical account would not anticipate structural outcomes such as Vietnam participating simultaneously in US-led and EU-led regulatory models, or Peru gaining higher centrality than the United States through mega-regional membership. By moving from the provision-level mapping that Elsig and Klotz (2021a, 2022) analyzed to modeling the network-level dynamics through which they spread, I connect two sides of the literature that have developed largely in parallel: one on the content and diffusion of digital trade rules, and another on the structural mechanisms that govern international cooperation more broadly (Hafner-Burton et al. 2009; Manger et al. 2012).

3 Data and Methods

3.1 Data Sources and Network Construction

The analysis draws primarily on the TAPED dataset (Burri and Polanco 2020), which codes digital provisions in PTAs across categories including e-commerce, data flow, and intellectual property. I match TAPED with bilateral gravity variables from the CEPII Gravity database (Conte et al. 2022) and EU membership indicators, yielding 5,133 dyad-PTA observations for the covariate models. TAPED records 404 PTAs entering into force between 2000 and 2022 across 173 countries; the three provision networks are built on the 382 of these that match the gravity and EU-membership covariates.

Using this data, I build three binary, undirected networks based on provision type: an *e-commerce network* (tie exists if two countries share any PTA containing at least one e-commerce provision, as coded in TAPED), a *data flow network* (at least one data flow provision), and an *intellectual property network* (at least one IP provision). Networks are binary rather than weighted to facilitate the use of a SAOM framework: a PTA with a single e-commerce provision is therefore treated identically to one with more provisions⁴. They are also cumulative: once a PTA enters into force, the tie persists over time, reflecting the durable nature of trade agreements. I also verify that no ties dissolve between consecutive periods across all provision types. EU member states appear as individual nodes despite collective negotiation, and I address this adding the “both EU members” homophily effect, to minimize convergence and misspecification by units issues. Plurilateral agreements (CPTPP, RCEP, EU-EPAs) create large cliques in the network, a feature the models accommodate but do not separately decompose.

The choice to disaggregate by provision type rather than building a single “digital provisions” network is substantively motivated in the theory. E-commerce, data flow, and IP provisions respond to different governance logics and involve different stakeholder coalitions. An aggre-

⁴SAOM results should therefore be read modeling the presence or absence of provision-type ties. In Appendix A I compare the correlations between binary centrality rankings vs. weighted rankings: the degree and eigenvector orderings the analysis relies on are largely preserved, while betweenness, which depends on shortest paths, is more sensitive to weighting.

gate network would mask the provision-type heterogeneity that is one of this paper’s central contributions. Table 1 summarizes the three networks at the end of the observation period (2022).

Table 1: Network Summary (2022)

Network	Countries	Ties	Share of Possible Ties	Clustering	Avg. Partners
E-Commerce	157	2296	18.7%	0.301	29.2
Data Flow	113	1480	23.4%	0.209	26.2
IP	138	2121	22.4%	0.210	30.7

Note: Clustering: share of connected triples that are fully closed. Cumulative binary undirected networks from TAPED (2000–2022).

3.2 Country Typology

To categorize states that might be playing different roles in how digital governance gets distributed, I propose a typology of four measured roles, Rulemaker, Adopter, Broker, and Peripheral, plus the EU as a structurally distinct bloc, based on provision-level novelty and structural network position. The four measured categories represent analytically useful groupings along a continuous spectrum of rulemaking influence, grounded on actual participation in digital governance agency as well as qualitative expectations from the literature; the EU enters on a separate institutional axis (collective negotiation) and is therefore evaluated apart from the measured roles.

The first category, the **EU Bloc**, is defined by institutional membership: EU member states negotiate trade agreements collectively through the European Commission, which generates nearly-identical network positions for all twenty-seven members regardless of individual regulatory capacity. The role of rulemaker for the EU will thus be evaluated separately.

Rulemakers are identified by demonstrated authorship of novel provisions rather than just network position: structural prominence and provision authorship diverge sharply, with the highest-novelty country sitting only at eigenvector rank 49 (Section 4.1). I measure agency directly, through three instruments that each capture template authorship through a different data-generating process. The first is a first-introduction novelty score, which tracks which

countries placed each of 114 individual digital governance provisions into a PTA before anyone else, across all agreements in the full TAPED dataset (Burri and Polanco 2020). The score assigns declining influence to successive introducers through a $1/\text{rank}$ decay and halves the credit of junior partners, those whose GDP is more than five times smaller than the agreement’s largest economy, so that economic leverage is not mistaken for design agency, as pinpointed by Elsig and Klotz (2022). The second is a time-ordered text-similarity measure, which rewards a country only when a later treaty closely reproduces the provision pattern of an earlier agreement it had already signed: first-mover design is thus valued more than copy-pasting. The third is a diffusion measure, which counts how often a country returns to negotiate later agreements that carry provisions it had itself first introduced, scaled by the number of provisions it authored, capturing sustained pushing for similar templates rather than one-off participation⁵.

I thus identify **Rulemakers** following two steps. First, I have a screening process based on the novelty score instrument: a non-EU country qualifies only if its first-introduction score reaches the top ten of the 125 non-EU authors, which removes states that co-signed others’ templates without writing novel provisions of their own (the exact cut-point and its construction are reported in Appendix B). Second, I focus on authorship as the substantive role of countries in the rulemaker group: a country that passes the screening process must also rank in the top eleven on at least one of the other two measures, text authorship or active diffusion. Nine countries fit this set of rules: the United States, Singapore, Australia, Chile, Japan, Canada, New Zealand, South Korea, and the United Kingdom.

The rule is not preemptively fitted to a roster of actors the literature already names, and the two diverge in informative ways. It keeps New Zealand, which a qualitative analysis of the scholarship does not conventionally treat as a norm leader (Jones et al. 2024), on its measured authorship and its founding role in the Digital Economy Partnership Agreement. It drops Mexico, a case the second step is built to identify: Mexico passes the first screening but ranks only twelfth on both other measures, confirming its smaller role as part of NAFTA negotiations. Six of the nine countries are independently identified as digital-

⁵Appendix B provides in-depth information about how each of these measures is constructed, as well as sensitivity tests to justify each choice.

trade template designers (the United States, Singapore, Australia, Japan, Canada, and the United Kingdom), and the other three are founders or early acceders to the Digital Economy Partnership Agreement.

The remaining countries are classified by median splits on two structural measures drawn from Burt (1992)’s framework: betweenness centrality⁶ and structural constraint⁷. **Brokers** bridge multiple clusters under constraint (above-median betweenness, above-median constraint); **Adopters** participate broadly with low structural dependence (below-median constraint); **Peripherals** have constrained positions and are relatively less important (below-median betweenness, above-median constraint). Table 2 reports the classification criteria with sample countries by category.

Table 2: Country Typology Classification Criteria

Category	Classification Rule	N	Examples
EU Bloc	EU membership	27	DEU, FRA, NLD, and 24 others
Rulemaker	Novelty screen (top-10, novelty score ≥ 23.66), then confirmed top-11 on ≥ 1 of the other 2 measures	9	USA, SGP, AUS, CHL, JPN, CAN, NZL, KOR, GBR
Broker	Betweenness $\geq$ median AND constraint $\geq$ median	19	IND, TUR, ISR, JOR, BRA
Adopter	Constraint $<$ median	56	VNM, PER, CHN, MEX, COL, CRI
Peripheral	Betweenness $<$ median AND constraint $\geq$ median	54	RUS, SAU, KAZ, most small economies

Note: Classification follows priority ordering: EU membership first, then Rulemaker (novelty screen, then confirmed on at least one of the other two measures), Broker, Adopter, Peripheral. Construction and sensitivity in Appendix B.

My proposed classification describes *structural roles*, rather than definitive attributions of agency: it identifies countries whose network position, entry timing, and capacity are consistent with template design. As external validation, most of the nine Rulemakers are the larger economy in the majority of their digital PTA partnerships, consistent with Elsig and Klotz (2022)’s finding that trade interests and domestic digital policy capacity at the initiator stage predict provision diffusion. The clearest exception is Singapore, which is smaller than many of its partners but is universally recognized as a template designer (Burri and Polanco 2020; Elsig and Klotz 2021a; Jones et al. 2024). Simultaneously, the novelty threshold allows us to exclude potentially non-designing economies. This includes Vietnam, Colombia, Peru, and

⁶Brokerage (betweenness centrality) counts the fraction of all shortest paths between pairs of countries that pass through a given country, normalized by the total number of these pairs. Countries with high betweenness are viewed as hubs, controlling information (possibly template flows) between clusters even if they are not themselves in the densest part of the network

⁷Burt’s constraint (showcasing structural autonomy) measures the extent to which a country’s partners are themselves densely interconnected. Constraint is computed as $C_i = \sum_j (p_{ij} + \sum_{q \neq i, j} p_{iq} \cdot p_{qj})^2$, where p_{ij} is the proportion of country i ’s connections that go to country j . Low constraint means a country’s partners are dispersed across separate clusters. High constraint means a country’s connections are concentrated among partners that are also tied to each other, limiting its structural independence.

China. The final list of countries is also consistent with alternative measurement strategies and thresholds (see Appendix B).

3.3 Network Analysis

To describe each actor’s position, characterize network centralities and growth, and ultimately build the country typology, I compute three sets of network-level and node-level measures for each provision-type network, annually from 2000 to 2022. These measures provide a first, descriptive account of the networks:

Centrality measures capture different aspects of structural position, as we specified while describing the typologies: *degree centrality* counts a country’s direct partners; *betweenness centrality* measures how often a country lies on shortest paths between others; and *eigenvector centrality*⁸ weights connections by the importance of partners, capturing embeddedness in the network core rather than pure connectivity (Hafner-Burton et al. 2009).

Structural constraint (Burt 1992) complements these centrality measures by capturing the *redundancy* of a country’s connections. A country whose partners are densely interconnected has high constraint (i.e., limited brokerage opportunity) while one whose partners are dispersed across separate clusters has low constraint and greater structural autonomy.

Network-level statistics (density, clustering coefficient, cross-network overlap) track aggregate structural evolution and can justify estimating separate models for each provision type rather than a single aggregate network; the cross-network overlap test reported in Appendix C (Table C1) supports estimating the three networks separately.

To provide information about network-related mechanisms, I further model network evolution using the Stochastic Actor-Oriented Model implemented in RSiena (Snijders 2001; Snijders et al. 2010). The SAOM treats tie formation as a multinomial choice: a focal country compares all potential tie changes and selects the one that most improves its “objective function” given the current network configuration (Block et al. 2019), which ends up mapping onto our theories of PTA formation where negotiating with one partner consumes resources

⁸While not used to define different typologies, eigenvector centrality is useful to understand the importance of a node (state) relatively to other nodes of the network.

unavailable for others.

I specify three observation waves at 2008, 2013, and 2020. Networks use only tie creation, reflecting the cumulative structure. The evaluation function includes two structural effects and four covariate effects. The structural effects are *triadic closure*, which captures the tendency to form ties that complete triangles, and *indirect connection closure* (the number of pairs at distance 2), which captures the tendency to form ties with countries two steps away (the structural arbitrage mechanism identified by Manger et al. (2012)); following Snijders (2023), I control for the simpler distance-2 configuration before interpreting the more complex transitive triads. The covariate effects are *economic capacity* (log GDP, ego effect) and *economic similarity* (GDP similarity), which capture hierarchy and homophily respectively; *both EU members* (homophily), which captures the EU’s collective negotiation pattern; and *geographic distance* (standardized), which captures proximity effects.

I estimate separate models for each provision type, allowing for direct comparison of how structural mechanisms work across the different domains I theorized, and report coefficients alongside approximate odds ratios in the text for interpretability (full odds-ratio table in Appendix D). The specification includes a model without indirect connection closure (used as baseline); the comparison between specifications is reported in Appendix E, while I run a Goodness of Fit check following Lospinoso and Snijders (2019) in Appendix F. An alternative wave specification (2008/2015/2022) was run as a robustness check: 17 of 18 effects maintain the same sign and 14 show full agreement (Appendix G).

4 Results

4.1 Network Structure and Country Roles

4.1.1 Network Growth and Visualization

All three networks have grown substantially since 2000 (Figure 1; Figure 2 below shows the e-commerce network at each SAOM observation wave, and Appendix H the three provision networks in 2022). By 2022, the e-commerce network encompasses 157 countries connected

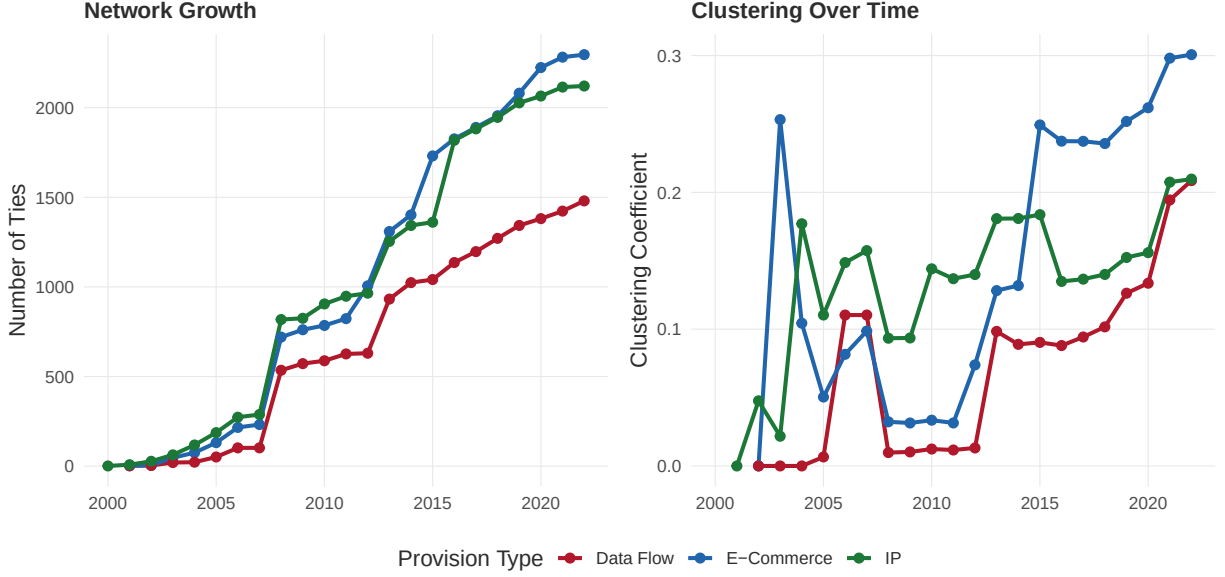


Figure 1: Evolution of Digital Provision Networks, 2000–2022. Left: cumulative number of ties. Right: clustering coefficient (triadic closure)

by 2,296 ties, the IP network covers 138 countries with 2,121 ties, and the data flow network, the smallest but fastest-growing, reaches 113 countries with 1,480 ties (Table 1). Data flow’s rapid expansion since 2015 may coincide with growing attention to cross-border data governance following the Snowden revelations and the GDPR (2016–2018), though attributing causality to specific events is beyond this paper’s scope. Clustering coefficients have risen from near zero to approximately 0.21–0.30 by 2022, indicating growing triadic closure (representing the probability for two countries that share a common PTA partner to eventually form a direct tie themselves) across all three networks.

Network centralization indices add further details to the story (Appendix I). In 2005, the data flow network was sharply more centralized (degree centrality = 0.62) than e-commerce (0.37) or IP (0.35), reflecting a governance domain concentrated around a few early hub countries. By 2022, degree centrality had converged across types, as mega-agreements and a generally denser set of PTAs distributed connectivity more broadly. Betweenness centralization, however, remains lower in the IP network (0.10) than in e-commerce (0.15) or data (0.15), suggesting that brokerage power is more diffuse in IP. This provides an initial characterization of IP governance as a relatively decentralized network built upon the common

foundation established by TRIPS.

As a sample visual representation, Figure 2 traces the e-commerce network across the three observation waves. The network becomes sharply more dense after 2013, as the mega-regional agreements draw spoke countries into the broader structure: the Rulemakers occupy bridging positions between otherwise separate groups, while the highly-central Adopters cluster around the growing core through mega-regional ties rather than through provisions they authored. Appendix H shows the same typology across all three provision networks in 2022.

Evolution of the E-Commerce Provision Network

Observation waves: 2008, 2013, 2020 | cumulative tie formation

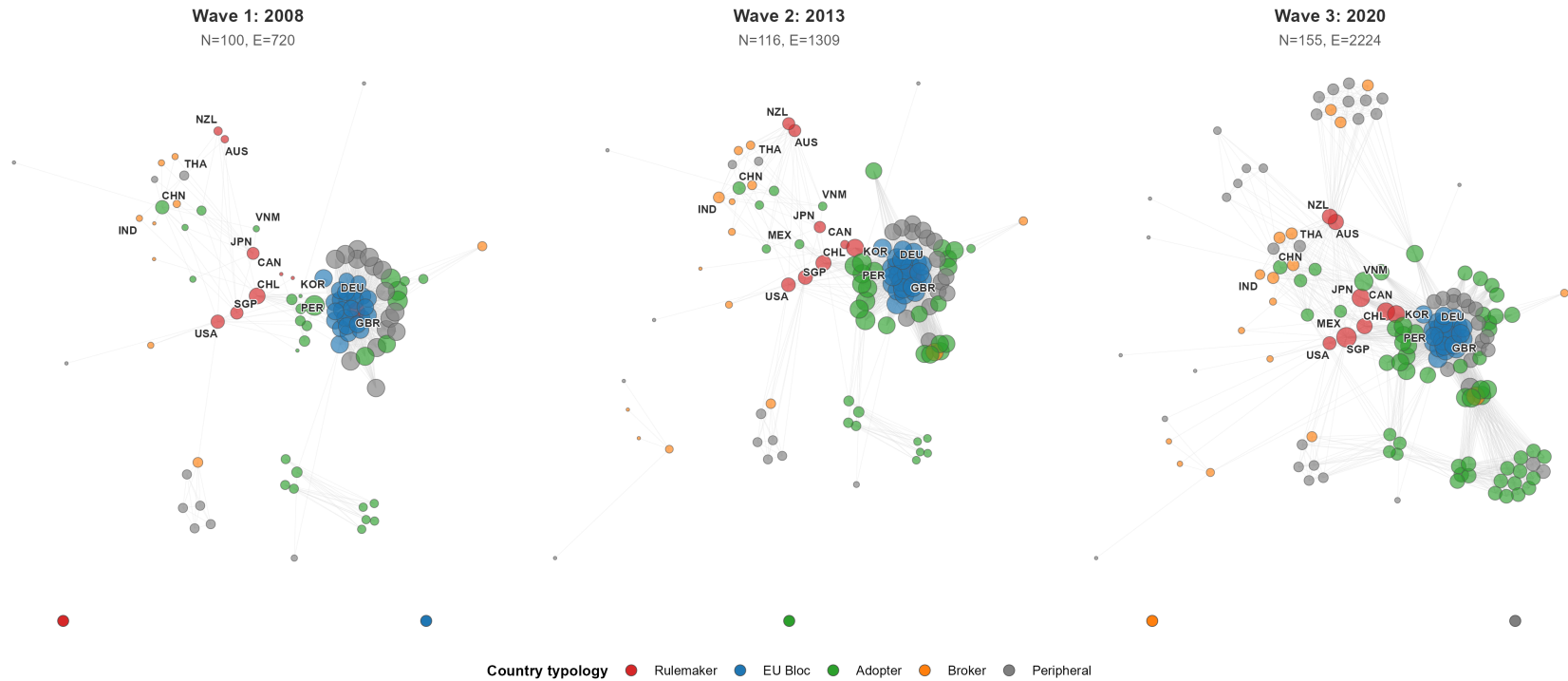


Figure 2: The e-commerce provision network at the three SAOM observation waves (2008, 2013, 2020). Node color denotes typology category; node size is proportional to the number of partners.

4.1.2 Country Roles

Before turning to the SAOM results, I characterize the structural positions that the models will explain.

Table 3: Most Connected Countries (E-Commerce Network, 2022)

Country	Partners	Betweenness	Structural Importance
South Korea	52	0.000	1.000
Singapore	61	0.013	0.985
Canada	46	0.013	0.964
Peru	48	0.025	0.943
Japan	49	0.013	0.926
Colombia	44	0.000	0.839
United Kingdom	98	0.236	0.804
Vietnam	53	0.000	0.801
Costa Rica	44	0.000	0.643
Honduras	40	0.000	0.620
United States	22	0.002	0.450
China	28	0.027	0.377

Note: Structural importance: eigenvector centrality. EU members receive near-identical scores due to collective negotiation. The US and China are included for reference even if outside the top 10.

Table 3 illustrates the e-commerce network as an example; similar patterns hold across all three provision types. EU member states receive nearly identical centrality scores due to their unique position as a regional bloc. South Korea and Singapore consistently rank among the highest in structural importance: in e-commerce, South Korea has the highest eigenvector centrality (1.00) and Singapore the second (0.99); in IP, South Korea ranks second (0.93) and Singapore fourth (0.88); in data, both remain in the top tier (SGP = 0.81, KOR = 0.80). Singapore stands out further for its brokerage role, with betweenness of 0.15 in data and 0.11 in IP, among the highest for non-EU countries across all networks.

The United States, by contrast, has moderate eigenvector centrality across all three networks (0.45 in e-commerce, 0.26 in data, 0.46 in IP) despite being widely recognized as a template designer. Its influence operates through deep, high-provision bilateral agreements (KORUS, USMCA) rather than network breadth, with only 22 direct e-commerce partners compared

to Singapore’s 61. The US does, however, show relatively higher betweenness in data (0.06), consistent with its role as initiator of data flow liberalization provisions. Peru and Vietnam both score higher than the United States in eigenvector centrality across networks, probably attributable to mega-regional memberships, which embed them in dense clusters of well-connected countries.

China presents yet another interesting profile: moderate eigenvector centrality across all three networks (0.38 in e-commerce, 0.23 in data, 0.35 in IP) but consistently notable betweenness. China’s moderate eigenvector but high betweenness reflects a targeted PTA strategy (Baccini et al. 2015) that connects otherwise separate clusters rather than embedding China as one of the most central players. This runs counter to the “digital empires” framing (Bradford 2023) that positions China alongside the US and EU as a co-equal rule-maker in digital governance: China is a significant *Adopter* that bridges clusters without designing the provisions that flow through those connections.

Zooming out and observing the various centrality measures, it is clear that there is a certain degree of asymmetry between structural prominence and provision authorship. Outside the EU, several of the most central nodes by eigenvector centrality (Peru, Colombia, Vietnam, and Costa Rica) are Adopters. Conversely, the United States holds the highest provision novelty score but sits at eigenvector rank 49. Network centrality alone rewards dense treaty participation over the actual design of provisions. Appendix J makes this distinction clear: when EU and mega-regional ties are stripped out, the centrality of these Adopters collapses, while the United States emerges as the most central actor in the e-commerce and IP networks and gains a prominent position in the data network.

4.1.3 Digital Governance Rulemakers

My typology classifications reveal what centrality-based rankings alone cannot capture. Figure 3 plots structural constraint against eigenvector centrality, and Table 4 reports the underlying scores for selected countries (the complete classification of all countries appears in Appendix K, Table K1). The plot is instructive precisely because neither axis draws the line between Rulemakers and the most prominent Ruletakers. Several Adopters sit at

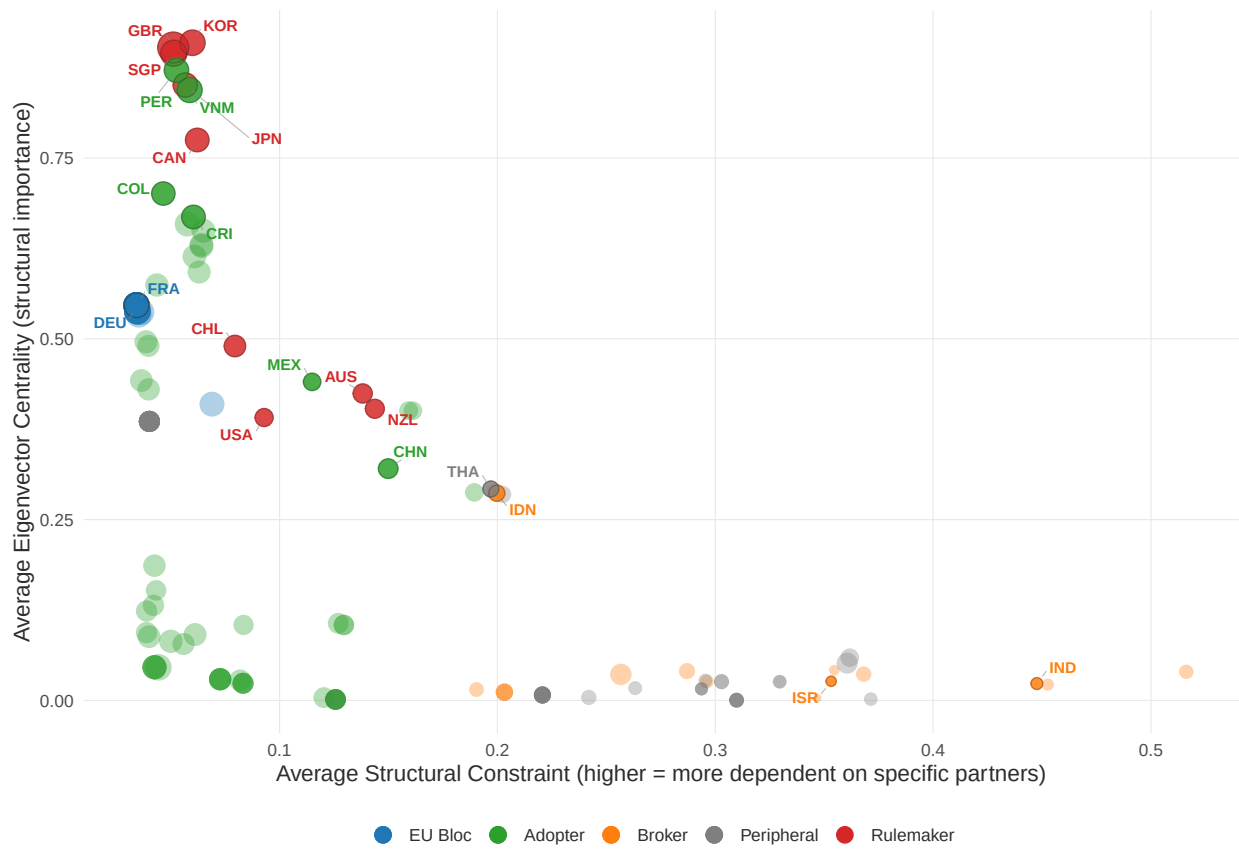


Figure 3: Country Typology by Eigenvector Centrality and Structural Constraint.

Table 4: Country Typology in Digital Provision Networks (2022)

ISO3	Country	Category	Eigenvector	Constraint	Betweenness	Entry Year
KOR	South Korea	Rulemaker	0.909	0.060	0.025	2004
GBR	United Kingdom	Rulemaker	0.903	0.051	0.204	2003
SGP	Singapore	Rulemaker	0.895	0.051	0.091	2001
JPN	Japan	Rulemaker	0.851	0.057	0.015	2002
CAN	Canada	Rulemaker	0.775	0.062	0.005	2002
CHL	Chile	Rulemaker	0.490	0.080	0.047	2003
AUS	Australia	Rulemaker	0.424	0.138	0.010	2003
NZL	New Zealand	Rulemaker	0.403	0.144	0.003	2001
USA	United States	Rulemaker	0.391	0.093	0.020	2001
DEU	Germany	EU Bloc	0.546	0.034	0.007	2003
FRA	France	EU Bloc	0.546	0.034	0.007	2003
NLD	Netherlands	EU Bloc	0.546	0.034	0.007	2003
PER	Peru	Adopter	0.871	0.053	0.010	2006
VNM	Vietnam	Adopter	0.844	0.059	0.018	2001
COL	Colombia	Adopter	0.701	0.047	0.020	2005
CRI	Costa Rica	Adopter	0.668	0.060	0.003	2002
MEX	Mexico	Adopter	0.440	0.115	0.028	2000
CHN	China	Adopter	0.320	0.150	0.041	2003
CHE	Switzerland	Adopter	0.104	0.083	0.023	2002
IDN	Indonesia	Broker	0.287	0.200	0.053	2003
IND	India	Broker	0.023	0.448	0.046	2004
THA	Thailand	Peripheral	0.292	0.197	0.000	2003

Note: Metrics averaged across e-commerce, data, and IP networks. Constraint: higher values indicate more structural dependence on specific partners. Entry year: earliest across all three provision types.

the very top of the eigenvector distribution: Peru (0.871) ranks above Japan (0.851), and Vietnam (0.844) ranks above Canada (0.775), with Colombia (0.701) and Costa Rica (0.668) also among the most central nodes. The constraint axis does not separate them either, since these Adopters sometimes carry constraint as low as the Rulemakers themselves; Peru at 0.053 is essentially indistinguishable from Singapore and the United Kingdom at 0.051. Despite these constraints, network centrality measures still allow us to position each category into substantively homogeneous groups, which entail similar degrees of structural constraint and centrality.

Looking at specific countries, Singapore is the clearest Rulemaker. Its first-introduction

novelty score is the second-highest of any country, behind only the United States, reflecting its role as one of the most prolific originators of digital governance provisions; it also entered very early (2001), carries high betweenness across networks (0.091, second only to the United Kingdom among Rulemakers), and holds high centrality across all provision types, though that centrality corroborates its prominence rather than grounding its classification. Burri (2020) and Elsig and Klotz (2021a) identify Singapore as a leading template designer in digital trade, and Jones (2024) provides detailed evidence that Singapore has acted as a “norm entrepreneur” in the post-2018 period, deliberately designing agreements like the DEPA as open, modular templates intended for broader adoption. The United States qualifies on the strength of the highest novelty score of any country, even though its eigenvector centrality is only moderate (0.391); its influence operates through deep bilateral templates (KORUS, US-Singapore, USMCA) and brokerage (betweenness 0.020) rather than network breadth, a pattern consistent with Elsig and Klotz’s (2022) finding that the US is the dominant initiator of digital trade provisions, even if those provisions do not themselves diffuse more readily than others’. Australia and New Zealand similarly carry only moderate eigenvector centrality (0.424 and 0.403), yet they remain well above the network median and qualify through high novelty scores (ranks 3 and 7). Japan, South Korea, Canada, and Chile are the remaining members, each combining high novelty scores with early entry, economic capacity, and genuine brokerage. The United Kingdom’s classification may partly reflect its post-Brexit bilateral agreement program, which largely rolled over provisions from EU agreements (Elsig and Klotz 2021a), though its novelty score (rank 9) provides independent support for template contribution.

From the perspective of other “rulemakers”, India exemplifies the powerful Broker: high betweenness paired with a structurally constrained position (constraint 0.448) that reflects a targeted rather than comprehensive PTA strategy (Baccini et al. 2015), bridging clusters without dominating any single one. China occupies a similar bridging position, but with structural constraint below the median it is classified an Adopter rather than a Broker; either way, it authors few of the provisions that travel through it. Peru is an Adopter with high centrality (eigenvector 0.871), gained through CPTPP and bilateral agreements. Vietnam is also an instructive case. Despite very early entry (2001, in the IP network), very

high eigenvector centrality (0.844), and low constraint (0.059), it is classified as an Adopter because its centrality derives from membership in CPTPP, RCEP, and the EVFTA rather than from authorship. The template-copying mechanism documented by Allee and Elsig (2019) and Peacock (2019) explains how countries like Vietnam gain structural prominence without contributing (on paper) to design of these provisions. Costa Rica, El Salvador, and other CAFTA-DR members present a similar pattern: high centrality gained from a US-designed template they adopted.

By combining provision-level novelty scores with structural network position, the proposed typology helps distinguish countries whose structural role is consistent with template design from those whose centrality is borrowed from well-connected agreements. Indeed, *structural position is informative but insufficient as an indicator of rulemaking agency*. Chile illustrates a related case: smaller than most of its partners, it nonetheless holds the third-highest betweenness among Rulemakers (0.047, behind only the UK and Singapore) and co-founded the Trans-Pacific Strategic Economic Partnership (P4, signed 2005, entered into force 2006) alongside New Zealand, Singapore, and Brunei, the agreement that became the template for the TPP and subsequently the CPTPP. Its classification as a Rulemaker is based on brokerage and demonstrated agency rather than small economic size.

4.1.4 First-Mover Advantage

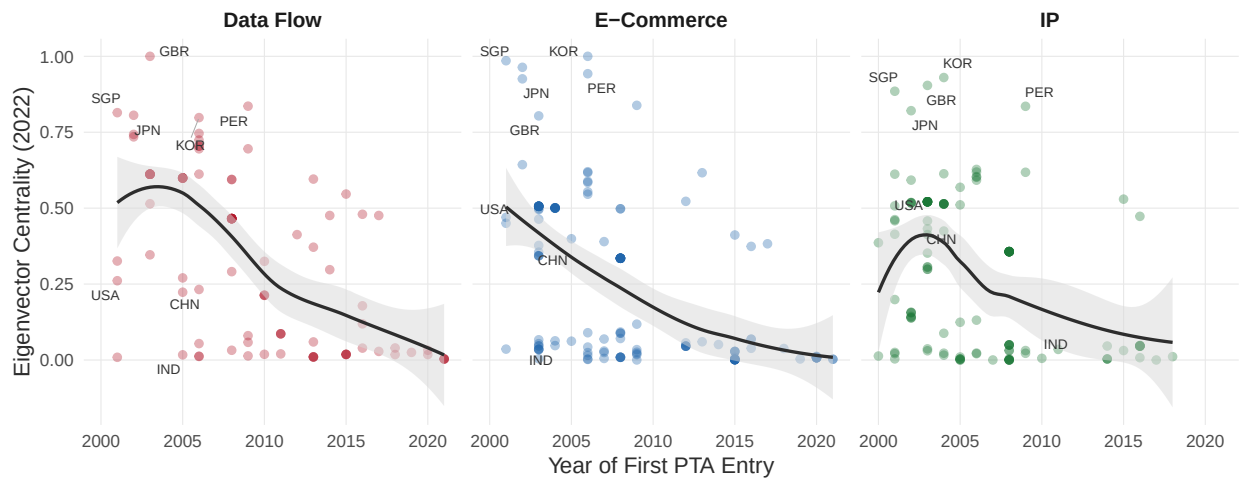


Figure 4: Entry Year vs. 2022 Eigenvector Centrality by Provision Type.

Countries that entered digital provision networks by 2005 seem to occupy persistently higher structural positions in 2022 (Figure 4). The temporal dimension matters for the rulemaker question: countries that signed the *first* PTAs containing digital provisions plausibly *designed* those provisions, while later entrants adopted existing templates. Early movers designed templates; latecomers adopted them. Elsig and Klotz (2022) provide direct evidence of this: roughly 60% of all the digital trade provisions the United States has introduced appear in a single early agreement, the US-Jordan FTA (2000). The relationship is strongest for e-commerce and weakest for IP, consistent with the IP network’s more universal, TRIPS-driven diffusion logic.

4.2 What Drives Digital Governance Tie Formation?

Table 5: What Drives Tie Formation? SAOM Results

Effect	E-Commerce	Data Flow	IP
Triadic closure	-0.078*** (0.015)	-0.092*** (0.030)	-0.473*** (0.088)
Indirect connection closure	-2.483*** (0.260)	-5.661*** (0.852)	-13.727 (225.329)
Geographic distance	0.078 (0.053)	1.453*** (0.319)	2.174*** (0.363)
Economic capacity (GDP)	1.483*** (0.321)	3.901*** (0.827)	1.093 (1.176)
Economic similarity	1.116** (0.547)	3.580*** (1.284)	-2.022 (1.321)
Both EU members	-3.509*** (0.292)	-11.247*** (1.557)	-15.744*** (0.604)

Note: Waves: 2008/2013/2020. Tie creation only. Distance standardized. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

I now turn to the SAOM results, which test the four hypotheses outlined in Section 2. Table 5 reports coefficients from separately estimated models for each provision type, and Figure 5 plots them with 95% confidence intervals.

4.2.1 Provision-type heterogeneity

My theory suggested that the three networks would follow distinct generative processes rather than a single diffusion logic. The cross-type coefficients substantiate this expectation: the mechanisms that predict tie formation in one network are insignificant or reversed in another, and the contrast is sharpest for the IP network, where economic hierarchy fails and structural arbitrage is left unidentified while distance strengthens. Geographic distance

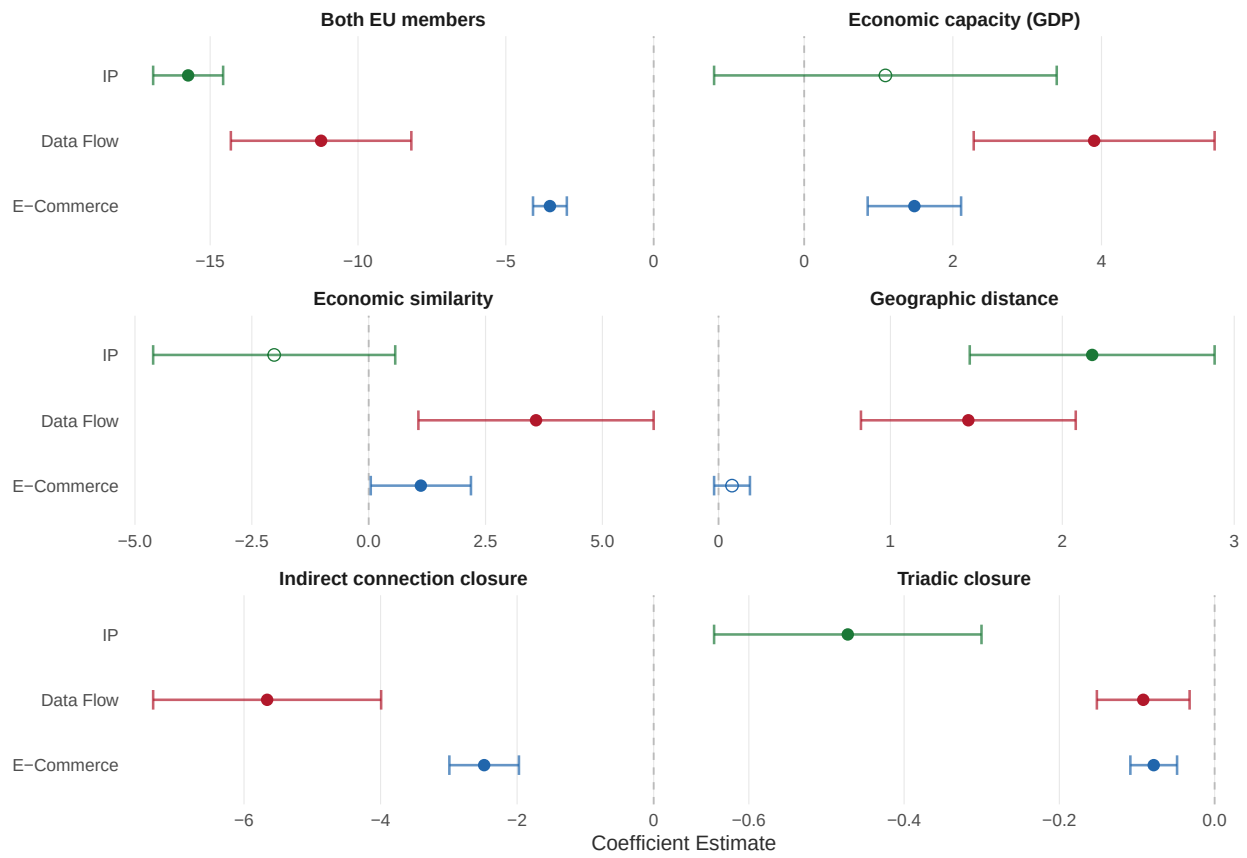


Figure 5: SAOM Coefficient Estimates with 95% Confidence Intervals. IP indirect connection closure is omitted from the panel (standard error = 225); its coefficient appears in Table 5.

shows no independent effect for e-commerce but is strongly positive for data and IP: distant countries are *more* likely to form data and IP ties, consistent with greater regulatory uncertainty between distant partners motivating explicit governance provisions (Peressotti 2025). The EU constraint shows progressively stronger external orientation as provision content becomes more regulatory. The most striking pattern is the IP network’s divergence from the other two: GDP effects are non-significant, distance is strongly positive, and structural arbitrage is unidentified. If partner characteristics do not matter as much, following the TRIPS-Plus literature IP provisions appear to diffuse through US-led pressure, rather than through the economic hierarchy that governs e-commerce and data governance networks.

4.2.2 Structural arbitrage, not generic clustering (H1)

The indirect connection closure effect is strongly negative for e-commerce (-2.48) and data (-5.66): countries with many partners-of-partners with whom they lack a direct tie experience structural pressure to close those gaps. In other words, spoke countries in hub-and-spoke configurations form direct ties to eliminate the hub’s intermediary advantage, extending the structural arbitrage logic of Manger et al. (2012) to digital governance. For the IP network, the indirect-connection-closure effect is unidentified: its standard error is roughly sixteen times the point estimate, so the primary model carries almost no information about structural arbitrage in IP. The contrast with e-commerce and data is thus one of identifiability as much as magnitude and, if irrelevant, would be consistent with TRIPS-Plus provisions proliferating through a more universal logic in which hub dependence plays little role. Further, once indirect connection closure is controlled for, triadic closure reverses sign to negative across all three networks. Countries form ties to bridge structural gaps, but once that structural-arbitrage motive is accounted for, they show no further preference for forming triangles. These results support H1: instead of generic clustering, structural arbitrage drives tie formation in the e-commerce and data networks, while leaving the mechanism unobserved for IP.

4.2.3 Economic hierarchy varies by provision type (H2)

Two distinct GDP mechanisms are at work. The *GDP ego effect* captures hierarchy: wealthier countries form more ties regardless of partner size. For data governance, this effect is the strongest (coefficient = 3.90, corresponding to approximately 49 times the odds per unit of log GDP), making data the most hierarchy-driven domain. The e-commerce effect is smaller but still substantial (1.48, approximately 4.4 times the odds). For IP, as I already addressed, GDP is non-significant after structural controls. H2 is supported: the hierarchy gradient runs from data (strongest) through e-commerce to IP (absent).

4.2.4 Economic similarity (H3)

The *GDP similarity effect* captures homophily: whether countries of similar economic size prefer each other. After controlling for indirect connection closure, GDP similarity turns positive for e-commerce and data, reversing the negative coefficient obtained without structural controls. For IP, it remains negative but non-significant. This result is substantively important: countries of similar economic size are more likely to form digital governance ties in e-commerce and data. H3 is partially supported: economic similarity promotes tie formation in e-commerce and data as predicted, while the IP result is in the predicted direction but non-significant.

4.2.5 EU structural constraint and the Brussels Effect (H4)

The EU’s institutional structure (collective negotiation through the European Commission, inability to form internal PTAs) produces a consistent structural peculiarity, coming from the fact that the “both EU members” coefficient is negative across all types. Part of this is mechanical, since EU members cannot form bilateral PTAs with each other, but the provision-type gradient could show something substantive: the EU’s external orientation intensifies as provision content becomes more regulatory (−3.5 for market-access e-commerce, −11.2 for data, −15.7 for enforcement-heavy IP), potentially reflecting deeper EU regulatory harmonization in data protection and IP. Further, using the TAPED subcategory coding, I classified each provision as either liberalizing (market-opening measures such as national

treatment, free cross-border data flow, and prohibitions on data localization) or constraining (regulatory measures such as data protection requirements, consumer protection, and general exceptions), and constructed a separate network for each orientation (Appendix L). The two networks are structurally near-identical: the same country pairs form ties whether the provisions liberalize digital markets or impose regulatory constraints. Constraining-only ties, however, concentrate among the EU’s partner countries (for example Albania, Algeria, Serbia, and ACP states), so the EU exports regulatory constraints to partners that receive no offsetting market-access provisions, a pattern consistent with the Brussels Effect. H4 is supported: EU member states form digital governance ties predominantly with non-EU partners, and the external orientation intensifies from e-commerce to IP.

5 Discussion

5.1 Rulemakers, Ruletakers, and Provision Heterogeneity

The SAOM results and the country typology are mutually reinforcing; reading them together reveals how macro-level positions emerge from micro-level forces. Strip out the SAOM, and the typology describes positions without identifying the forces that generate them; strip out the typology, and the SAOM identifies forces without anchoring them to the actors whose agency this paper seeks to explain. For example, structural arbitrage, the dominant driver of tie formation in e-commerce and data, yields the exact low-constraint positions that characterize Rulemakers. Similarly, the GDP ego effect concentrates ties around high-income hubs, explaining why the Rulemaker category fills with wealthy, early-entering economies.⁹

Still, results make it clear that hierarchy structures digital trade governance, yet the pattern is more uneven than the “digital empires” framework appears to suggest. Singapore emerges as the most consistent Rulemaker (analytically, and substantively, based on the literature): it combines early entry, high centrality across all provision types, among the highest betweenness across networks, and low structural constraint. Jones (2024) provides independent validation defining it as a norm entrepreneur in digital trade since the US stepped back

⁹Network endogeneity is still a concern. Despite mitigations through SAOM-based simulations, omitted-variable bias cannot be fully ruled out.

from active rulemaking after 2018, showing that in all 14 instances of a new issue being introduced into digital trade agreements Singapore was a fundamental party. The United States is also a Rulemaker, but its influence operates through deep bilateral templates rather than network breadth, which explains why it does not rank as strongly in eigenvector centrality despite being widely recognized as a leading digital rulemaker (Elsig and Klotz 2022). The EU, operating as a collective regulatory exporter with the strongest structural presence in the SAOM specification, acts as a de facto rulemaker despite its unique position.

The central finding for the rulemaker question is that **centrality, as a standalone measure, is a poor proxy for agency**. Peru and Vietnam have higher eigenvector centrality than the United States in the e-commerce network, not because they design provisions, but because CPTPP membership embeds them in a dense cluster of well-connected countries. By incorporating provision-level novelty and structural brokerage alongside centrality and entry timing, my proposed multi-dimensional typology separates adoption density from design agency. Vietnam is classified as an Adopter despite its exceptional network metrics; its centrality comes from agreements designed by others. The typology cannot fully resolve this distinction without provision-level content analysis, but pairing it to studies like Elsig and Klotz (2022) and Jones et al. (2024) substantially improves metrics based on centrality rankings. Further, the SAOM results help us understand the mechanisms through which different country types operate. Rulemakers like Singapore sit inside a double-edged structural arbitrage dynamic: as hubs, they create the indirect connections that generate pressure for spoke-to-spoke tie formation, and the same arbitrage that lets spokes bypass a hub also carries the hub’s templates outward, since new ties inherit templates from prior agreements. Structural centrality and regulatory influence can thus move in opposite directions, an effect that echoes the “weaponized interdependence” logic of Farrell and Newman (2019). Brokers like India occupy high-betweenness positions connecting otherwise disconnected clusters, but their constrained positions limit their template-setting capacity and appear to be relatively less influential than the literature gives them credit for.

As for adopters, they seem to gain centrality through participation rather than initiative. China bridges clusters, yet its low structural constraint places it among autonomous Adopters

rather than Brokers. More importantly, its first-introduction novelty score falls below the necessary threshold to make it a full-fledged rulemaker. This reading aligns with Elsig and Klotz (2022), who note China has “mostly adopted existing commitments.” Its substantial regulatory power goes instead through domestic legislation, like the Personal Information Protection Law, and limited WTO engagement, channels that PTA network analysis simply does not capture. Mexico represents another archetype in the Adopter category. Despite early entry and a high novelty score, it fails in being considered a rulemaker based on text-authorship or diffusion measures. It remains a NAFTA-model adopter rather than an independent designer.

Finally, the EU and the intellectual property network underline deeper structural dynamics. The negative EU effect across all three networks confirms that member states form ties predominantly outside the bloc. The EU’s outward digital focus intensifies as provisions become more regulatory; IP diffuses from a common TRIPS baseline rather than through hub elimination. Meanwhile, the IP network diverges sharply from e-commerce and data. GDP hierarchy effects are non-significant, and structural arbitrage is unidentified. As such, even in the IP network, the novelty-identified Rulemakers are still structurally distinctive, which is what one would expect if their status reflects provision authorship rather than a single network mechanism.

5.2 Implications for Digital Governance

The main implication for how international organizations govern new technologies is that digital rulemaking, rather than being ignored, appears to have found a new governance vehicle in the form of PTAs. The fragmented regime complex for digital governance (Alter and Meunier 2009) is structured by identifiable mechanisms like structural arbitrage, economic hierarchy, and institutional constraints, which systematically advantage some actors over others. The provision-type heterogeneity finding is important for institutional design in particular, since e-commerce, data, and IP provisions follow different structural logics; a single multilateral agreement would need to accommodate fundamentally different governance dynamics within one framework (which underlines the complexity of conducting negotiations

with such a breadth in scope). The persistent difficulty of WTO e-commerce negotiations may substantially reflect structural incompatibility across governance domains (Burri 2022). The rise of standalone Digital Economy Agreements represents the next phase of this change, moving governance even further toward streamlined, designer-led templates.

The distributive stakes of this shift are also important. Whoever writes the early, low-stakes template sets the defaults that later entrants inherit (Castle 2023). Since rules spread through hub-spoke elimination rather than centralized negotiation, hub templates become the standard of choice rather than one option among many. This raises an important question: whether the WTO or comparable bodies can ever recapture governance authority that has already shifted to an alternative network more capable of adapting to the ever-changing nature of technology.

6 Conclusion

In this article I argue that the distinction between rulemakers and ruletakers in digital governance is produced by the interaction of entry timing, structural arbitrage, and provision type heterogeneity. Looking at the structural network components of digital provisions as part of trade agreements, using SAOM analysis of three provision-specific networks, and combining them to propose a multi-dimensional country typology, I find that e-commerce and data governance networks grow through the closure of indirect connections, while IP provisions follow a distinct, more universal logic. I also find that formation dynamics vary qualitatively across provision types, and that the rulemaker/ruletaker distinction requires attention to entry timing, structural constraints, and provision content, not just centrality alone.

The typology, distinguishing four measured roles (nine Rulemakers, Adopters, Brokers, and Peripheral countries) plus the EU as a structurally distinct bloc, provides a more discriminating framework for understanding power in digital governance. Singapore occupies the most consistently central position; the EU’s influence operates through collective external orientation rather than individual centrality; and many ostensibly “central” countries, including

Vietnam, Peru, and the CAFTA-DR members, are adopters whose structural position is borrowed from mega agreements they did not design. Among them, China is one of the most striking examples: in the context of PTA digital governance, rather than being a co-equal rulemaker together with other major economies such as the US and the EU, it plays a more limited role connecting different hubs of the PTA network. This runs counter to what would be expected according to Bradford (2023) and Aaronson and Leblond (2018).

For policymakers, the implication is that structural position in the PTA network is an unreliable guide to rulemaking influence if taken as a standalone measure. Entry timing matters: first-movers gain persistent advantages. Structural constraint is fundamental: unconstrained positions enable brokerage. And finally, provision content is important as well: the same network structure carries different regulatory content depending on the provision type. For scholars, the paper shows the value of provision-type disaggregation, of extending the structural-arbitrage mechanism from general PTA networks to provision-specific digital ones, and of multi-dimensional typologies that distinguish rulemakers from rule-takers within international economic governance.

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Appendix

A. Binary vs. Weighted Centrality

Table A1: Binary vs. Weighted Centrality Rank Correlations (2022)

Network	Eigenvector	Degree	Betweenness
E-Commerce	0.72	0.78	0.26
Data Flow	0.95	0.95	0.66
IP	0.85	0.83	0.36

Note: Spearman rank correlation of binary centrality rankings used throughout the paper against weighted rankings, in which edge weights equal provision counts, for the 2022 networks.

The binary degree and eigenvector orderings the analysis relies on are largely preserved under provision-count weighting, while betweenness, which depends on shortest paths, is more sensitive to weighting.

B. Typology: Construction, Sensitivity, and the Pre-Screen Alternate

B.1 The three agency instruments

The status of Rulemaker depends on three country-level instruments, each measuring template authorship through a different process.

1. *First-introduction novelty*. For each substantive digital-trade provision, I rank every PTA carrying it by entry-into-force year, assigning time-rank r (first agreement $r = 1$) and novelty credit $1/r$. Each signatory receives a share after two adjustments: on EU-negotiated treaties each member's share is divided by the number of EU signatories, and a junior partner, whose log GDP is more than log 5 below the largest partner's, has its share multiplied by 0.5. A country's score sums its adjusted credits across all provisions and agreements.

2. *Time-ordered text authorship*. Each PTA is a binary vector over the provision indicators. For every agreement pair with Jaccard overlap at least 0.70, I order them in time and credit only common signatories already party to the earlier agreement, with weight $J \times 0.5^{\Delta t/5}$, so a five-year gap halves the weight.

3. *Active diffusion*. A country's diffusion score is the number of later agreements it signs that carry one of its own first-introduced provisions, divided by the number of provisions it first introduced; a country that first-introduced no provision scores zero.

B.2 Classification rule and cut-points

A non-EU country is a Rulemaker if it passes the novelty threshold and is confirmed on a second measure. The threshold requires a first-introduction novelty score of at least 23.66, the top ten of the 125 non-EU authors; 23.66 is the midpoint of the step between the tenth-ranked author (Mexico, 25.04) and the eleventh (Colombia, 22.29). Confirmation requires a top-eleven rank on at least one of the other two instruments, text authorship or active diffusion. Confirmation on either instrument, rather than both, reflects that the two measures capture distinct and non-substitutable authorship channels (verbatim text propagation and repeated re-introduction of a country's own provisions): a designer may

exert influence primarily through one channel, so requiring agreement on both would screen out genuine authors rather than sharpen the category. The confirmation band is set one rank wider than the screen (top eleven rather than top ten) to accommodate Chile, the single country whose measured status turns on this choice: Chile ranks eleventh on both corroborating instruments, and its inclusion rests on independent evidence of design agency (co-founder of the P4 template and the third-highest betweenness among Rulemakers, Section 4.1.3) rather than on the band alone. The three rulemaker categories form a two-axis grid on betweenness centrality and Burt’s structural constraint, both split at their medians over the classified rulemaker pool (0.0020 and 0.1904): constraint below the median yields an Adopter; above the median, above-median betweenness yields a Broker and below-median betweenness a Peripheral country.

B.3 Sensitivity

Membership is consistent across different metrics, and its *exclusions* are guaranteed by construction: because a Rulemaker must clear the novelty screening process, the central but non-designing economies (Vietnam, Colombia, Peru, China) cannot enter at any rank threshold from $K = 8$ to $K = 15$, and every threshold in the interval (22.29, 25.485) returns the identical nine-country set. Among high-novelty authors, eight of the nine are invariant across top- K values from 9 to 12; Chile is the one country whose status turns on the $K = 10/11$ boundary, and Mexico the only non-member that ever enters, at $K \geq 12$.

				threshold = 23.66 (operating)				screen = 40.77 (k-means, strict)				
country				K=9	K=10	K=11	K=12	K=9	K=10	K=11	K=12	
USA	SGP	AUS	JPN	in	in	in	in	in	in	in	in	core (all cells)
CAN	NZL	GBR	KOR	in	in	in	in	.	.	.	.	chosen nine
CHL				.	.	in	in	.	.	in	in	margin (K=10/11)
MEX				.	.	.	in	.	.	.	.	margin (K=11/12)
VNM	COL	PER	CHN	.	.	.	.	.	.	.	.	excluded (screen veto)

B.4 The pre-screening alternative typology

I present an alternative rule to define Rulemakers, which drops the screening process and keeps only the second step, top- K on at least two of the three instruments. This measure differs from the primary specification by a single substitution: it admits Vietnam, which

passes the second step on its early CPTPP and ASEAN co-signature despite ranking eighteenth on its own novelty, and it drops Chile, whose two corroborating instruments tie at the $K = 11$ boundary. The main, more constraining typology choice appears to be consistent and substantially similar to this alternative version (showcased for completeness).

K	Primary (with screen)	Pre-screen alternate	Countries the screen removes
9	8 designers	9	Vietnam
10	8 designers	9	Vietnam
11 (operating)	9 (adds Chile)	11	Colombia, Vietnam
12	10 (adds Mexico)	13	Colombia, Peru, Vietnam

C. Cross-Network Overlap

Table C1: Cross-Network Overlap (QAP Test, 2022)

Network Pair	Jaccard Overlap	Correlation	p-value
E-Commerce / Data	0.644	0.775	< 0.001
E-Commerce / IP	0.618	0.719	< 0.001
Data / IP	0.616	0.743	< 0.001

Note: Jaccard: proportion of shared ties among all ties in either network. QAP: Quadratic Assignment Procedure with 1,000 permutations. Moderate overlap (0.62–0.64) which justifies separate estimation.

D. Odds Ratios

Table D1: SAOM Coefficients and Odds Ratios

effect_label	estimate_E-Commerce	estimate_Data Flow	estimate_IP	or_str_E-Commerce	or_str_Data Flow	or_str_IP
Triadic closure	-0.0784567	-0.0919467	-0.4725368	0.92	0.91	0.62
Indirect connection closure	-2.4828526	-5.6612937	-13.7268078	0.08	0.00	0.00
Geographic distance	0.0784487	1.4532368	2.1736812	1.08	4.28	8.79
Economic capacity (GDP)	1.4825218	3.9012973	1.0927044	4.40	49.47	2.98
Economic similarity	1.1163253	3.5802747	-2.0217048	3.05	35.88	0.13
Both EU members	-3.5086009	-11.2473712	-15.7443022	0.03	0.00	0.00

Note: Odds ratios = $\exp(\text{coefficient})$. An OR of 4.40 for GDP ego in e-commerce means a one-unit increase in log GDP multiplies the odds of tie formation by approximately 4.4.

E. Specification Comparison: With and Without Indirect-Connection Closure

Table E1: Coefficient Comparison, Primary vs. Baseline Specification

effect_label	Primary_ecommerce	Primary_data	Primary_ip	Baseline_ecommerce	Baseline_data	Baseline_ip
Triadic closure	-0.078 (0.015)	-0.092 (0.030)	-0.473 (0.088)	0.044 (0.004)	0.076 (0.011)	0.057 (0.008)
Indirect connection closure	-2.483 (0.260)	-5.661 (0.852)	-13.727 (225.329)	NA	NA	NA
Geographic distance	0.078 (0.053)	1.453 (0.319)	2.174 (0.363)	-1.294 (0.629)	25.158 (1.524)	33.915 (1.768)
Economic capacity (GDP)	1.483 (0.321)	3.901 (0.827)	1.093 (1.176)	1.601 (0.241)	15.907 (3.292)	2.868 (0.678)
Economic similarity	1.116 (0.547)	3.580 (1.284)	-2.022 (1.321)	-0.272 (0.273)	-0.819 (0.273)	-2.521 (0.336)
Both EU members	-3.509 (0.292)	-11.247 (1.557)	-15.744 (0.604)	-1.450 (0.070)	-2.195 (0.112)	-2.960 (0.109)

Note: The primary specification adds indirect-connection closure and standardizes geographic distance; the baseline omits indirect-connection closure.

F. Goodness of Fit

Table F1: Goodness-of-fit Mahalanobis-distance test p-values (primary specification).

Network	Degree	Triad census	Geodesic
E-commerce	< 0.001	< 0.001	< 0.001
Data flow	< 0.001	< 0.001	< 0.001
IP	< 0.001	< 0.001	< 0.001

Note: Monte Carlo Mahalanobis-distance test (Lospinoso and Snijders 2019), 3,000 simulations; auxiliary statistics not used in estimation. All three networks converged (maximum convergence ratios below 0.25).

I assess goodness of fit for the primary specification with the Monte Carlo Mahalanobis-distance test of Lospinoso and Snijders (2019), simulating three auxiliary statistics not used in estimation: the degree distribution, the triad census, and the geodesic-distance distribution. The test rejects exact fit on every statistic in all three networks: the networks are cumulative and estimated on tie creation only, so the model can add ties but never remove them, which mechanically constrains the degree and distance distributions it can reproduce.

G. Wave Sensitivity

Table G1: Wave Sensitivity Analysis

Effect	Provision	Primary (2008/2013/2020)	Alternative waves (2008/2015/2022)	Agreement
Triadic closure	E-Commerce	-0.078*	-0.026*	Full
Triadic closure	Data Flow	-0.092*	0.043	No
Triadic closure	IP	-0.473*	-0.094*	Full
Indirect connection closure	E-Commerce	-2.483*	-1.843*	Full
Indirect connection closure	Data Flow	-5.661*	-5.561*	Full
Indirect connection closure	IP	-13.727	-7.704*	Sign only
Geographic distance	E-Commerce	0.078	0.003	Full
Geographic distance	Data Flow	1.453*	0.914*	Full
Geographic distance	IP	2.174*	1.310*	Full
Economic capacity (GDP)	E-Commerce	1.483*	2.605*	Full
Economic capacity (GDP)	Data Flow	3.901*	7.662*	Full
Economic capacity (GDP)	IP	1.093	2.215*	Sign only
Economic similarity	E-Commerce	1.116*	2.127*	Full
Economic similarity	Data Flow	3.580*	4.169*	Full
Economic similarity	IP	-2.022	-1.656*	Sign only
Both EU members	E-Commerce	-3.509*	-2.953*	Full
Both EU members	Data Flow	-11.247*	-13.221*	Full
Both EU members	IP	-15.744*	-11.003*	Full

Note: Of 18 effect comparisons, 17 maintain the same sign and 14 show full agreement. Core effects are fully robust.

H. The Three Provision Networks, 2022

Digital Provision Networks, 2022 Nodes colored by typology, sized by degree

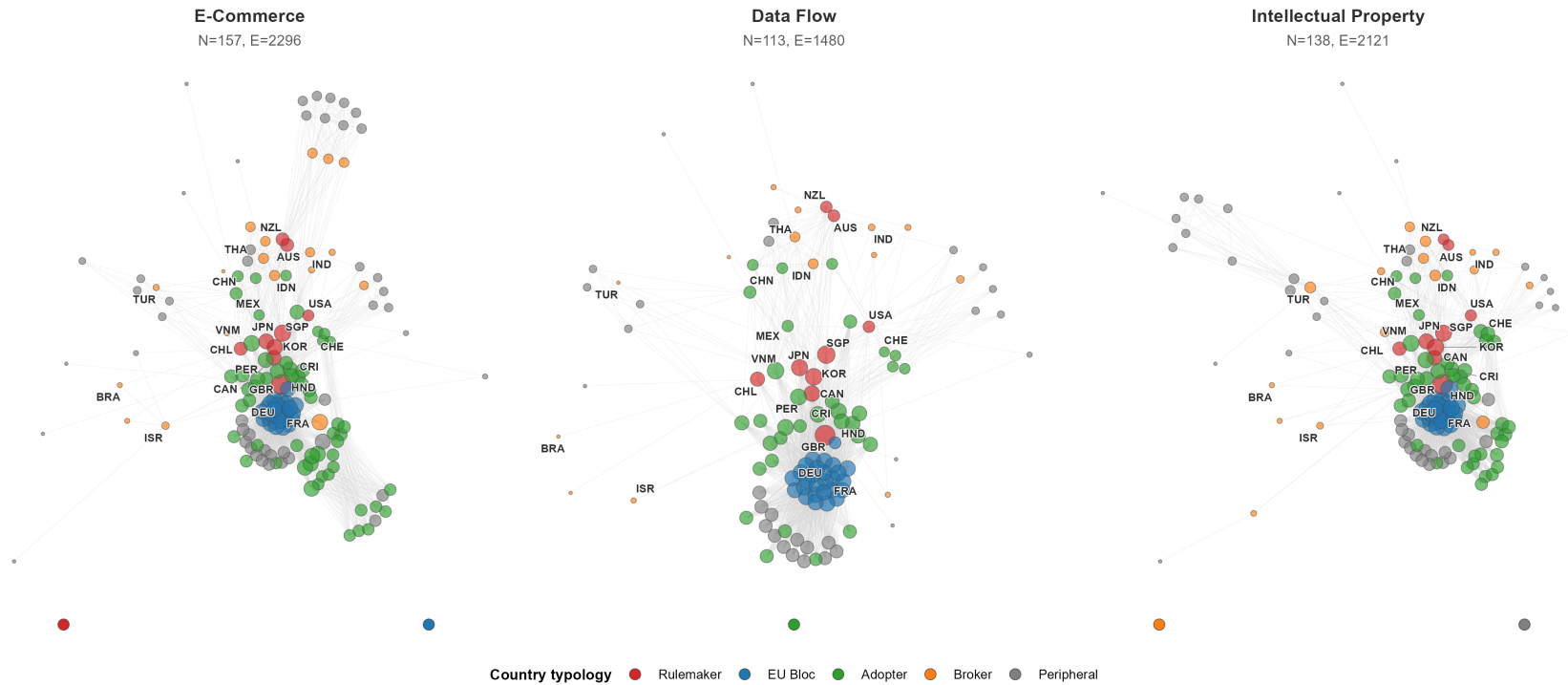


Figure H1: The three digital provision networks (e-commerce, data flow, intellectual property) in 2022. Node color denotes typology category; node size is proportional to the number of partners.

I. Network Centralization Indices

Table I1: Network Centralization Indices by Provision Type and Year

Year	Provision Type	Degree	Betweenness	Eigenvector
2003				
2003	Data Flow	0.580	0.414	0.869
2003	E-Commerce	0.311	0.124	0.900
2003	IP	0.243	0.230	0.821
2005				
2005	Data Flow	0.618	0.506	0.828
2005	E-Commerce	0.370	0.350	0.796
2005	IP	0.352	0.235	0.826
2008				
2008	Data Flow	0.258	0.317	0.402
2008	E-Commerce	0.188	0.249	0.577
2008	IP	0.184	0.152	0.630
2013				
2013	Data Flow	0.217	0.173	0.478
2013	E-Commerce	0.178	0.115	0.476
2013	IP	0.203	0.148	0.575
2020				
2020	Data Flow	0.256	0.163	0.466
2020	E-Commerce	0.210	0.104	0.527
2020	IP	0.242	0.085	0.577
2022				
2022	Data Flow	0.463	0.155	0.656
2022	E-Commerce	0.441	0.151	0.708
2022	IP	0.455	0.104	0.669

Note: Centralization ranges from 0 (perfectly equal) to 1 (one node dominates). Calculated from cumulative binary undirected networks (TAPED, 2000–2022).

J. Counterfactual Centrality

This counterfactual is a hard test for the typology: if the Rulemakers hold their positions once the dense EU cluster and the CPTPP and RCEP mega-agreements are stripped away, their centrality reflects provision design rather than network density. I rebuild the 2022 networks after removing the ties created by CPTPP, RCEP, and 34 EU-negotiated agreements, dropping a dyad's tie only when every digital-provision PTA it shares falls in this set, which removes 62% of e-commerce ties, 82% of data ties, and 75% of IP ties; the Rulemakers come through it largely intact. The United States, United Kingdom, Singapore, Australia, South Korea, and New Zealand all remain in the counterfactual top fifteen across networks, and the United States, which belongs to none of the removed agreements, climbs from the periphery to the top of the centrality ranking in e-commerce and IP and into the top tier in data (eigenvector rank 4); the Rulemakers that did join the mega-deals, namely Canada, Japan, and Korea, shed most of their ties but stay well above the collapsing Adopters, with Korea holding a top rank while Canada and Japan settle into the mid-table. The EU Bloc tells the opposite story, as 15 of 16 EU members tested fall to zero ties in e-commerce and all 16 in data and IP, which confirms that their position is an artifact of collective negotiation rather than individual activity; the two highest-centrality Adopters fall just as hard, with Vietnam dropping from rank 8 to 25 in e-commerce and from 1 to 23 in IP, and Peru from 4 to 14 in e-commerce, since their centrality runs almost entirely through CPTPP, RCEP, and, for Vietnam, the EU-Vietnam agreement. That drop is specific to them, since Adopters whose ties run through bilateral rather than mega-regional agreements, such as Costa Rica and Colombia, are far less affected.

K. Country Typology: Full Classification

Table K1: Full Country Typology Classification (2022).

Category	ISO3	Country	Eigenvector	Betweenness	Constraint	Entry Year
Rulemaker						
Rulemaker	AUS	Australia	0.424	0.0101	0.138	2003
Rulemaker	CAN	Canada	0.775	0.0052	0.062	2002
Rulemaker	CHL	Chile	0.490	0.0469	0.080	2003
Rulemaker	GBR	United Kingdom	0.903	0.2044	0.051	2003
Rulemaker	JPN	Japan	0.851	0.0151	0.057	2002
Rulemaker	KOR	South Korea	0.909	0.0246	0.060	2004
Rulemaker	NZL	New Zealand	0.403	0.0034	0.144	2001
Rulemaker	SGP	Singapore	0.895	0.0912	0.051	2001
Rulemaker	USA	United States	0.391	0.0205	0.093	2001
Adopter						
Adopter	AGO	Angola	0.024	0.0044	0.083	2015
Adopter	ALB	Albania	0.078	0.0219	0.056	2002
Adopter	ARM	Armenia	0.496	0.0010	0.039	2015
Adopter	BDI	Burundi	0.001	0.0001	0.126	2015
Adopter	BIH	Bosnia & Herzegovina	0.186	0.0052	0.043	2001
Adopter	BRN	Brunei	0.401	0.0000	0.161	2003
Adopter	BWA	Botswana	0.029	0.0209	0.073	2008
Adopter	CHE	Switzerland	0.104	0.0233	0.083	2002
Adopter	CHN	China	0.320	0.0411	0.150	2003
Adopter	CIV	Côte d'Ivoire	0.124	0.0000	0.039	2016
Adopter	CMR	Cameroon	0.131	0.0029	0.042	2014
Adopter	COD	Congo - Kinshasa	0.024	0.0044	0.083	2015
Adopter	COL	Colombia	0.701	0.0197	0.047	2005
Adopter	COM	Comoros	0.046	0.0313	0.044	2012
Adopter	CRI	Costa Rica	0.668	0.0030	0.060	2002
Adopter	DJI	Djibouti	0.001	0.0001	0.126	2015
Adopter	DOM	Dominican Republic	0.592	0.0000	0.063	2006
Adopter	DZA	Algeria	0.152	0.0000	0.043	2005
Adopter	ECU	Ecuador	0.574	0.0149	0.044	2005
Adopter	ERI	Eritrea	0.001	0.0001	0.126	2015
Adopter	GEO	Georgia	0.442	0.0177	0.037	2016
Adopter	GHA	Ghana	0.094	0.0000	0.039	2016
Adopter	GTM	Guatemala	0.630	0.0020	0.064	2006
Adopter	HND	Honduras	0.649	0.0000	0.065	2006
Adopter	ISL	Iceland	0.107	0.0871	0.127	2002
Adopter	KEN	Kenya	0.004	0.0082	0.120	2015
Adopter	LBY	Libya	0.001	0.0001	0.126	2015

Table K1: Full Country Typology Classification (2022). *(continued)*

Category	ISO3	Country	Eigenvector	Betweenness	Constraint	Entry Year
Adopter	LIE	Liechtenstein	0.104	0.0240	0.130	2002
Adopter	LSO	Lesotho	0.029	0.0209	0.073	2008
Adopter	MDA	Moldova	0.430	0.0013	0.040	2004
Adopter	MDG	Madagascar	0.046	0.0200	0.043	2012
Adopter	MEX	Mexico	0.440	0.0277	0.115	2000
Adopter	MNE	Montenegro	0.091	0.0610	0.061	2002
Adopter	MOZ	Mozambique	0.028	0.0416	0.082	2015
Adopter	MWI	Malawi	0.024	0.0044	0.083	2015
Adopter	MYS	Malaysia	0.401	0.0013	0.159	2003
Adopter	NAM	Namibia	0.029	0.0209	0.073	2008
Adopter	NIC	Nicaragua	0.614	0.0001	0.061	2006
Adopter	NOR	Norway	0.104	0.0240	0.130	2002
Adopter	PAN	Panama	0.659	0.0085	0.058	2004
Adopter	PER	Peru	0.871	0.0102	0.053	2006
Adopter	PHL	Philippines	0.288	0.0426	0.189	2003
Adopter	PNG	Papua New Guinea	0.088	0.0297	0.040	2009
Adopter	RWA	Rwanda	0.001	0.0001	0.126	2015
Adopter	SLV	El Salvador	0.628	0.0000	0.064	2002
Adopter	SRB	Serbia	0.082	0.0730	0.050	2002
Adopter	SWZ	Eswatini	0.029	0.0209	0.073	2008
Adopter	SYC	Seychelles	0.046	0.0200	0.043	2012
Adopter	TTO	Trinidad & Tobago	0.386	0.0000	0.040	2008
Adopter	TZA	Tanzania	0.024	0.0044	0.083	2015
Adopter	UGA	Uganda	0.001	0.0001	0.126	2015
Adopter	UKR	Ukraine	0.490	0.0222	0.040	2001
Adopter	VNM	Vietnam	0.844	0.0179	0.059	2001
Adopter	ZAF	South Africa	0.029	0.0209	0.073	2008
Adopter	ZMB	Zambia	0.046	0.0200	0.043	2012
Adopter	ZWE	Zimbabwe	0.046	0.0200	0.043	2012
Broker						
Broker	ARG	Argentina	0.015	0.0027	0.773	2005
Broker	BHR	Bahrain	0.026	0.0091	0.296	2003
Broker	BRA	Brazil	0.016	0.0077	0.781	2005
Broker	FJI	Fiji	0.011	0.0349	0.203	2020
Broker	HKG	Hong Kong SAR China	0.041	0.0377	0.287	2003
Broker	IDN	Indonesia	0.287	0.0526	0.200	2003
Broker	IND	India	0.023	0.0457	0.448	2004
Broker	ISR	Israel	0.026	0.0391	0.353	2000
Broker	JOR	Jordan	0.022	0.0597	0.453	2001

Table K1: Full Country Typology Classification (2022). *(continued)*

Category	ISO3	Country	Eigenvector	Betweenness	Constraint	Entry Year
Broker	LAO	Laos	0.288	0.0044	0.199	2003
Broker	MKD	North Macedonia	0.015	0.0087	0.190	2001
Broker	MMR	Myanmar (Burma)	0.036	0.0601	0.368	2003
Broker	MUS	Mauritius	0.036	0.0293	0.257	2012
Broker	SLB	Solomon Islands	0.011	0.0349	0.203	2020
Broker	TUN	Tunisia	0.005	0.0075	0.757	2005
Broker	TUR	Turkey	0.040	0.0799	0.516	2003
Broker	TWN	Taiwan	0.042	0.0023	0.355	2004
Broker	VEN	Venezuela	0.003	0.0227	0.347	2005
Broker	WSM	Samoa	0.011	0.0349	0.203	2020
EU Bloc						
EU Bloc	AUT	Austria	0.546	0.0074	0.034	2003
EU Bloc	BEL	Belgium	0.546	0.0074	0.034	2003
EU Bloc	BGR	Bulgaria	0.537	0.0117	0.037	2002
EU Bloc	CYP	Cyprus	0.538	0.0062	0.035	2004
EU Bloc	CZE	Czechia	0.538	0.0062	0.035	2004
EU Bloc	DEU	Germany	0.546	0.0074	0.034	2003
EU Bloc	DNK	Denmark	0.546	0.0737	0.034	2003
EU Bloc	ESP	Spain	0.546	0.0074	0.034	2003
EU Bloc	EST	Estonia	0.539	0.0062	0.035	2002
EU Bloc	FIN	Finland	0.546	0.0074	0.034	2003
EU Bloc	FRA	France	0.546	0.0074	0.034	2003
EU Bloc	GRC	Greece	0.546	0.0074	0.034	2003
EU Bloc	HRV	Croatia	0.410	0.0419	0.069	2001
EU Bloc	HUN	Hungary	0.538	0.0062	0.035	2004
EU Bloc	IRL	Ireland	0.546	0.0074	0.034	2003
EU Bloc	ITA	Italy	0.546	0.0074	0.034	2003
EU Bloc	LTU	Lithuania	0.539	0.0062	0.035	2002
EU Bloc	LUX	Luxembourg	0.546	0.0074	0.034	2003
EU Bloc	LVA	Latvia	0.540	0.0062	0.035	2002
EU Bloc	MLT	Malta	0.538	0.0062	0.035	2004
EU Bloc	NLD	Netherlands	0.546	0.0074	0.034	2003
EU Bloc	POL	Poland	0.538	0.0062	0.035	2004
EU Bloc	PRT	Portugal	0.546	0.0074	0.034	2003
EU Bloc	ROU	Romania	0.533	0.0115	0.035	2001
EU Bloc	SVK	Slovakia	0.538	0.0062	0.035	2004
EU Bloc	SVN	Slovenia	0.538	0.0062	0.035	2004
EU Bloc	SWE	Sweden	0.546	0.0074	0.034	2003
Peripheral						

Table K1: Full Country Typology Classification (2022). *(continued)*

Category	ISO3	Country	Eigenvector	Betweenness	Constraint	Entry Year
Peripheral	AFG	Afghanistan	0.000	0.0000	0.310	2008
Peripheral	ARE	United Arab Emirates	0.017	0.0009	0.263	2001
Peripheral	ATG	Antigua & Barbuda	0.386	0.0000	0.040	2008
Peripheral	AZE	Azerbaijan	0.000	0.0000	0.310	2008
Peripheral	BHS	Bahamas	0.386	0.0000	0.040	2008
Peripheral	BLR	Belarus	0.026	0.0010	0.330	2009
Peripheral	BLZ	Belize	0.386	0.0000	0.040	2008
Peripheral	BOL	Bolivia	0.003	0.0000	1.000	2006
Peripheral	BRB	Barbados	0.386	0.0000	0.040	2008
Peripheral	COK	Cook Islands	0.008	0.0000	0.221	2020
Peripheral	DMA	Dominica	0.386	0.0000	0.040	2008
Peripheral	EGY	Egypt	0.052	0.0000	0.361	2004
Peripheral	ETH	ETH	0.001	0.0001	0.126	2015
Peripheral	FRO	Faroe Islands	0.002	0.0000	0.507	2006
Peripheral	GRD	Grenada	0.386	0.0000	0.040	2008
Peripheral	GUY	Guyana	0.386	0.0000	0.040	2008
Peripheral	HTI	Haiti	0.386	0.0000	0.040	2008
Peripheral	IRN	Iran	0.002	0.0000	0.371	2008
Peripheral	JAM	Jamaica	0.386	0.0000	0.040	2008
Peripheral	KAZ	Kazakhstan	0.026	0.0015	0.303	2008
Peripheral	KGZ	Kyrgyzstan	0.026	0.0015	0.303	2008
Peripheral	KHM	Cambodia	0.285	0.0000	0.202	2003
Peripheral	KIR	Kiribati	0.008	0.0000	0.221	2020
Peripheral	KNA	St. Kitts & Nevis	0.386	0.0000	0.040	2008
Peripheral	KWT	Kuwait	0.016	0.0002	0.294	2003
Peripheral	LBN	Lebanon	0.059	0.0000	0.362	2006
Peripheral	LCA	St. Lucia	0.386	0.0000	0.040	2008
Peripheral	LKA	Sri Lanka	0.029	0.0000	1.000	2018
Peripheral	MAC	Macao SAR China	0.002	0.0000	1.000	2003
Peripheral	MAR	Morocco	0.017	0.0000	0.745	2006
Peripheral	MHL	Marshall Islands	0.008	0.0000	0.221	2020
Peripheral	MNG	Mongolia	0.028	0.0000	1.000	2016
Peripheral	NIU	Niue	0.008	0.0000	0.221	2020
Peripheral	NRU	Nauru	0.008	0.0000	0.221	2020
Peripheral	OMN	Oman	0.027	0.0002	0.295	2003
Peripheral	PAK	PAK	0.004	0.0016	0.242	2008
Peripheral	PLW	Palau	0.008	0.0000	0.221	2020
Peripheral	PRY	Paraguay	0.000	0.0000	1.000	2005
Peripheral	PSE	Palestinian Territories	0.001	0.0000	1.000	2005
Peripheral	QAT	Qatar	0.016	0.0002	0.294	2003

Table K1: Full Country Typology Classification (2022). *(continued)*

Category	ISO3	Country	Eigenvector	Betweenness	Constraint	Entry Year
Peripheral	RUS	Russia	0.026	0.0010	0.330	2015
Peripheral	SAU	Saudi Arabia	0.016	0.0002	0.294	2003
Peripheral	SDN	SDN	0.001	0.0001	0.126	2015
Peripheral	SUR	Suriname	0.386	0.0000	0.040	2008
Peripheral	SYR	Syria	0.000	0.0000	1.000	2007
Peripheral	THA	Thailand	0.292	0.0000	0.197	2003
Peripheral	TJK	Tajikistan	0.000	0.0000	0.310	2008
Peripheral	TKM	Turkmenistan	0.000	0.0000	0.310	2008
Peripheral	TON	Tonga	0.008	0.0000	0.221	2020
Peripheral	TUV	Tuvalu	0.008	0.0000	0.221	2020
Peripheral	URY	Uruguay	0.018	0.0000	0.767	2004
Peripheral	UZB	Uzbekistan	0.000	0.0000	0.310	2008
Peripheral	VCT	St. Vincent & Grenadines	0.386	0.0000	0.040	2008
Peripheral	VUT	Vanuatu	0.008	0.0000	0.221	2020

Eigenvector centrality, betweenness centrality, and Burt's constraint are averaged across all three provision-type networks (e-commerce, data flow, IP); entry year is the earliest year of participation in any digital provision PTA.

L. Liberalizing vs. Constraining Provision Networks

Table L1: Liberalizing vs. Constraining Provision Networks (2022)

	2022
Liberalizing ties	1,222
Constraining ties	1,491
Ties present in both networks	1,219
Liberalizing-only ties	3
Constraining-only ties	272
Constraining-only ties involving an EU member	222 (82%)

Note:

Ties classified by TAPED provision orientation; a dyad may carry liberalizing ties, constraining ties, or both. The two orientation networks share 1,219 of their ties. The EU-member share is computed over the constraining-only dyads.